

Project ZephyrZero: Super-Efficient
Transport Group Design Project. Flight
Dynamics and Control analysis

ZEPHYR ZERO



SUPER EFFICIENT TRANSPORT

Name: Pablo Egerique G.C. (SET 05)

CID: 02029946

Abstract

This report presents a preliminary study of the dynamic stability characteristics of the Super Efficient Transport aircraft from the project ZephyrZero; a computational analysis of said aircraft using VLM in the OpenVSP aerodynamic solver, VSPAERO , and a uniform triangle Galerkin panel method using flow5; and the preliminary design of the empennage of the aircraft, of T-tail structure, together with a preliminary sizing of a Vee-tail configuration investigating possible improvements in performance. The aircraft is found to have satisfactory dynamical stability modes, agreeing with trends from similar aircraft data; the software are able to predict to a reasonable degree of accuracy the stability characteristics of the aircraft and the Vee-tail is found to improve the performance and future studies are recommended.

Role Definitions and Project Members

Configuration Definition	Clarizze Cuntapay Cammy Nguyen Zong Junn Lee Jose Salazar Carrasco
Propulsion Systems	Jakub Titi Karthik Krishnan Sharwa Molla Nicholas Westwood Chi Him Chun
Aerodynamic Design	Jai Hyung Tommy Ko Angus Millar Mohammed Razib Haider Toha Oliver Fraser
Structural Design	Rakshansingh Hritik Parmessur Samuel Paterson Viraaaj Darbar Israr Mahmud
Flight Dynamics and Control	Pablo Egerique G. C. Rohan Kukreti Ruixin Yan Puqi Liang Haohua Howard Chen
Aircraft Safety and Ground Infrastructure	Aloysius Yang Vilok Avadhan Yang Jiao

Contents

List of Figures	iv
List of Tables	v
List of Symbols	vi
1 Introduction	1
2 Equations of motion and dynamic modes evaluation	1
2.1 The linearised aircraft equations of motion	1
2.1.1 Derivation procedure	1
2.1.2 Limitations of the analysis for the proposed configuration	1
2.2 Moments of inertia calculation	2
2.3 Longitudinal motion	2
2.4 Lateral motion	3
3 Computational modelling of the aircraft	3
3.1 Software selection	4
3.2 Analysis process in OpenVSP	4
3.2.1 Modelling the aircraft	4
3.2.2 Analysis methodology in VSPAERO	5
3.2.3 Mesh convergence study	6
3.3 Analysis process in flow5	7
3.3.1 Modelling the aircraft	7
3.3.2 Analysis methodology in flow5	7
3.3.3 Mesh convergence study	8
3.4 Evaluation of results	9
4 Empennage design	10
4.1 Empennage configuration decision	10
4.2 T-tail design	10
4.2.1 Horizontal tailplane sizing	10
4.2.2 Vertical tailplane sizing	11
4.3 Vee-tail design	11
5 Further work & conclusions	12
Appendix A: Supporting equations	15
Appendix B: OpenVSP model supporting data	16
Appendix C: flow5 model supporting data	20
Appendix D: Vee-tail data supporting figures	23

List of Figures

1	Dynamical modes' eigenvalues	3
2	Final model tested in OpenVSP	5
3	flow5 final model	7
4	flow5 mesh comparison between base mesh and final mesh	8
5	Model tested in OpenVSP	16
6	Clustering visualisation	16
7	Fuselage number of elements effects on the results convergence in OpenVSP	17
8	Wing number of elements effects on the results convergence in OpenVSP	18
9	Horizontal tailplane number of elements effects on the results convergence in OpenVSP	18
10	Truss number of elements effects on the results convergence in OpenVSP	19
11	Vertical tailplane number of elements effects on the results convergence in OpenVSP	19
12	Fuselage effects in aerodynamic calculations in flow5	20
13	flow5 mesh convergence study for different wing tessellation	21
14	flow5 mesh convergence study for different horizontal tailplane tessellation	21
15	flow5 mesh convergence study for different truss tessellation	22
16	flow5 mesh convergence study for different vertical tailplane tessellation	22
17	Vee-tail control surface design assumption, reproduced from [1]	23
18	Variation of stability parameters with Γ & b . Fixed $c_r = 3$ m	23
19	Variation of stability parameters with Γ & c_r . Fixed $b = 18$ m	24
20	Vee-tail aircraft preliminary design for the second iteration design	24

List of Tables

1	Inertial properties of the aircraft evaluated at the cruise stage	2
2	Longitudinal eigenvalues for the cruise segment conditions	2
3	Lateral eigenvalues for the cruise segment conditions	3
4	Final tessellation parameters for the OpenVSP model	6
5	Final mesh parameters for the flow5 model	8
6	Lateral derivatives comparison	9
7	Eigenvalue comparison (with the changes outlined in [2] applied)	9
8	Tailplane aerofoil selection process	11
9	Tailplane geometric parameters	11
10	Base tessellation parameters for the OpenVSP model	17
11	Base mesh parameters for the flow5 model	20

List of Symbols

Roman symbols

x	Body-fixed axis
y	Body-fixed axis
z	Body-fixed axis
X	force in the x -direction
Y	force in the y -direction
Z	force in the z -direction
U	x -component of velocity
V	y -component of velocity
W	z -component of velocity
L	moment about the x -axis (regarding stability) or lift (regarding aerodynamics)
M	moment about the y -axis (pitching moment)
N	moment about the z -axis (yawing moment)
p	roll rate
q	pitch rate
r	yaw rate
D	drag force
Ma	Mach number
g	gravitational constant
S_{ref}	wing reference area
$D_{f_{max}}$	maximum fuselage diameter
T_d	damped period
Re	Reynolds number
$\bar{c}$	mean aerodynamic chord
m	mass of the aircraft at the evaluated stage
$\mathcal{AR}$	Aspect ratio
I_{ij}	mass moment of inertia about the $x / y / z$ axis, represented by i and j
$\bar{V}_i$	volume coefficient of horizontal tailplane or vertical tailplane, represented by i
S_i	horizontal tailplane or vertical tailplane area, represented by i
l_i	horizontal tailplane or vertical tailplane moment arm, represented by i
b_i	span of wing or horizontal tailplane or vertical tailplane, represented by i
N	number of elements

Greek symbols

α	angle of attack
β	sideslip angle
ρ	density
μ	dynamic viscosity
ω_n	natural frequency
ζ	damping ratio
λ	taper ratio
Φ	roll angle
Ψ	yaw angle
Θ	pitch angle
Γ	circulation (regarding aerodynamics) or dihedral angle (regarding lifting surface design)
Λ	sweep angle

General notation convention

C_*	coefficient of *
$\bullet_*$	derivative of $\bullet$ with respect to *
$\bullet_e$	$\bullet$ evaluated at equilibrium conditions
$\dot{\bullet}$	time derivative of $\bullet$
$\bullet_{cg}$	x, y or z location (represented by $\bullet$) of the centre of gravity

1 Introduction

The study of the flight dynamics and control characteristics of an aircraft is a crucial part of the design process. This task has several aspects and requires many iterations, in conjunction with different speciality sub teams, to optimise the performance of the design, whilst guaranteeing its safety. This report presents an analysis of the dynamic modes of the aircraft by analysing the equations of motion of the latter; a computational modelling of the design in two different software, outlining the predicted parameters and carrying out a comparison with analytically obtained ones; and it also presents the design procedure followed for the design of the empennage structure of the aircraft, describing an optimised design of a T-tail and the preliminary sizing of a Vee-tail that fulfills the stability requirements and could result in a potential performance improvement, therefore laying the initial steps to be followed in future iterations.

2 Equations of motion and dynamic modes evaluation

2.1 The linearised aircraft equations of motion

2.1.1 Derivation procedure

In order to evaluate the dynamic characteristics of the aircraft it is necessary to derive a set of equations of motion that govern its dynamical behaviour. Due to the natural complexity of the system, once derived, the equations are linearised, therefore confining the validity of the analysis to a small range of perturbations, being this the first limitation of the analysis. Regardless, it is a very powerful tool to obtain an idea of the aircraft's behaviour in various stages of its flight regime. Additionally, the set of assumptions made during the derivation cause a slight loss of generality and a potential loss of accuracy in the results, which, in the case of a novel configuration such as that of the aircraft presented, may cause discrepancies between these results and those obtained from simulations. This will be examined in a later stage of this report.

The derivation of these equations has been carried out extensively in literature [3, 4, 5] using similar procedures and assumptions (although varying notation). The reference process chosen is outlined by Drela in [5], but it is the general method followed by other authors. When describing a moving body (translating and rotating) it is common to define two different sets of axis systems: "Earth" axes and "Body-fixed" axes, the latter are the ones used to carry out the analysis of the aircraft dynamics. This simplifies certain calculations, however, it brings some additional complications, namely the need of translating from one axis system to another (to, for example, translate the reference position of the body, generally expressed in "Earth" axes, to "Body-fixed" axes). This is done using a rotation matrix composed of the product of trigonometric relations of Euler angles (Eq. (9.4) in [5]).

Having established the above, the remaining tasks are to derive the kinematic and dynamic relations of the aircraft. This can be done by analysing the position and orientation rate of change of the aircraft (for kinematic) and applying Newton's second law and Euler's equation (for dynamic) to the system. By formulating all of the latter, it is possible to obtain 12 expressions, for which 12 states can be defined: the aircraft position in "Earth" axes (although these components are likely to have a minimal effect and will therefore be disregarded from the analysis [6]), the three Euler angles, and the 3 velocity components and 3 rotation rates of change in the "Body-fixed" axes system. By linearising the system through a Taylor expansion approximation, assuming small perturbations around the equilibrium state (e.g. trim position), the linearised system for the aircraft can be obtained. Additionally, a control term including the effects of control surfaces can be added to the system, however, for the purposes of the analysis outlined in this report this effect will not be considered. Further simplifications of the system include disregarding the effects of position in aerodynamic forces; the assumption of $\dot{U}$ & $\dot{V}$ derivatives having no effect since we are dealing with a heavier-than-air vehicle; and considering the $\dot{p}$, $\dot{q}$ & $\dot{r}$ derivatives to have a small effect (empirically) [6], therefore disregarding them. Finally, applying the Curie principle, which assumes that there will be no effect from asymmetric causes on symmetric states and vice versa, it is possible divide the obtained system into two systems of equations, one lateral and one longitudinal.

2.1.2 Limitations of the analysis for the proposed configuration

The assumptions and simplifications described above limit the accuracy of the analysis and may induce errors in the results produced. The linearisation of the system carries some potential errors with it. Firstly, the expressions are only applicable in the case of small disturbances. Additionally, it is assumed that the equilibrium wind velocity is zero and has therefore no effect on the equilibrium velocity of the aircraft and gyroscopic effects generated by spinning rotors are neglected [3]. Despite this, previous results have found the small perturbation approximation to give good predictions for the stability characteristics of aircraft, making it therefore a reasonable method to use [3]. In dividing the equations using the Curie principle, there are further assumptions made as outlined in [3]. In order to have pure longitudinal motions, it is assumed that the aircraft has an axis of symmetry and gyroscopic effects are neglected. In order to have pure lateral motions, additionally to previously mentioned assumptions, the aerodynamic force cross-coupling is neglected. It is worth mentioning that, if the equations are not linearised, there are several inertial cross-coupling terms, which may induce additional effects that have not been accounted for. The effect of the mentioned elements may impact the results, therefore increasing the disagreement between them and the computationally obtained ones, which will be presented and discussed in a later stage of this report.

2.2 Moments of inertia calculation

Before describing the calculations of the dynamical properties of the aircraft it is important to describe the methods used to estimate its inertial properties. These can be estimated using the Class II methods outlined by Roskam in [7]. By assuming I_{xy} & I_{yz} are zero, since the aircraft is symmetrical [7], the moments of inertia can be estimated through Eqs. 5 - 8, found in Appendix A. Additionally, the use of the software OpenVSP in the design process of the aircraft (which will be described later in this report) allowed to carry out a computational estimation of the inertial properties. After the mass properties of each component have been introduced, the OpenVSP software calculates the inertial properties by slicing the model into a certain number of slices. After checking the convergence of the results, the final values of the location of the center of gravity, the total mass, and the moments of inertia were estimated. By obtaining the mass distributions of the the different aircraft components from the SET01 (configuration definition) sub team and introducing them both in the analytical equations and the software, the values outlined in Table 1 were obtained.

	I_{xx}	I_{yy}	I_{zz}	I_{xz}
Analytical calculations (kgm ²)	4.709e+05	1.454e+06	1.767e+06	6.410e+04
OpenVSP (kgm ²)	4.867e+05	1.493e+06	1.821e+06	6.793e+04
Percentage difference (%)	3.287	2.620	3.010	5.8

Table 1: Inertial properties of the aircraft evaluated at the cruise stage

It can be observed that the results generally agree. Slight deviations can be seen both in the parameters seen in Table 1 and were also observed in the comparison of the centre of gravity and total mass estimation (of around 1.5% for both), which may be due to the slicing method used by the software. Despite this, the differences are not considered to be major. In order to keep consistency, the analytical results were calculated using the analytical inertial properties and similarly the computational calculations were carried out with the computational values. This is to be able to obtain a clear comparison of overall results produced by the different sources - computational and analytical - (which will be outlined in a later stage of this report). Further iterations should carry out a more in-depth computation of inertial properties.

2.3 Longitudinal motion

From the analysis described above, it is possible to introduce the linearised equations governing the longitudinal motion of the aircraft as follows:

$$\begin{bmatrix} m & -X_{\dot{W}} & 0 & 0 \\ 0 & m - Z_{\dot{W}} & 0 & 0 \\ 0 & -M_{\dot{W}} & I_{yy} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\Theta} \end{bmatrix} = \begin{bmatrix} X_U & X_W & X_q & -mg \cos \Theta_e \\ Z_U & Z_W & Z_q + mU_e & -mg \sin \Theta_e \\ M_U & M_W & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \Theta \end{bmatrix} \quad (1)$$

The system is composed of the longitudinal stability derivatives, inertial components, and the equilibrium values of the freestream velocity and the equilibrium pitch angle. To calculate the longitudinal behaviour of the aircraft the above ODE needs to be solved. By assuming a normal mode solution, the above can be transformed into an eigenvalue problem where the eigenvalues of the operator will characterise the long-term response. The inertial parameters can be obtained from the geometry and mass distribution of the aircraft, the stability derivatives are outlined in [8], and the equilibrium values should be inputted depending on the stage evaluated. Being cruise the main segment of the mission, the equilibrium values are used: $U_e = 221.39 \text{ ms}^{-1}$ & $\Theta_e = -0.7^\circ$. With these parameters, it is possible to obtain an estimate of the eigenvalues characterising the aircraft's dynamic modes.

	Eigenvalue	ω_n (1/s)	ζ	T_d (s)	Time to double / half amplitude (s)
Phugoid	$0.008 \pm 0.040i$	0.041	-	153.248	86.643
SPPO	$-0.590 \pm 0.774i$	0.973	0.606	8.118	1.170

Table 2: Longitudinal eigenvalues for the cruise segment conditions

There are two longitudinal dynamical modes and they are characterised by the parameters in Table 2 and can be seen in Figure 1a for different flight stages. The following analysis will be carried out for the cruise condition values, although the eigenvalues for the Take-Off segment and Landing segment follow a similar trend apart from the Phugoid mode, which is stable in these two segments as opposed to the cruise segment. The Phugoid mode is a low-frequency mode characterised by a slow sinusoidal motion governed by the aircraft's x & z velocity components whilst the incidence angle remains virtually constant [4, 6]. The mode description agrees with the values obtained for this aircraft. In this particular case, the mode is unstable, however, this is generally accepted since the frequency is very low and therefore the time to half amplitude is large enough to give the pilot the opportunity to adjust the aircraft controls. This however would not be the case if the SPPO (Short Period Pitching Oscillation) had a positive real part. This mode is characterised by a sudden change in pitch following a longitudinal disturbance whilst the

aircraft velocity remains generally constant [4] and is governed by the pitch stability derivative (M_w , or C_{m_α}), which is generally found to be negative [5]. Again, like it was the case for the Phugoid mode, the values found for the aircraft agree with the expected trends.

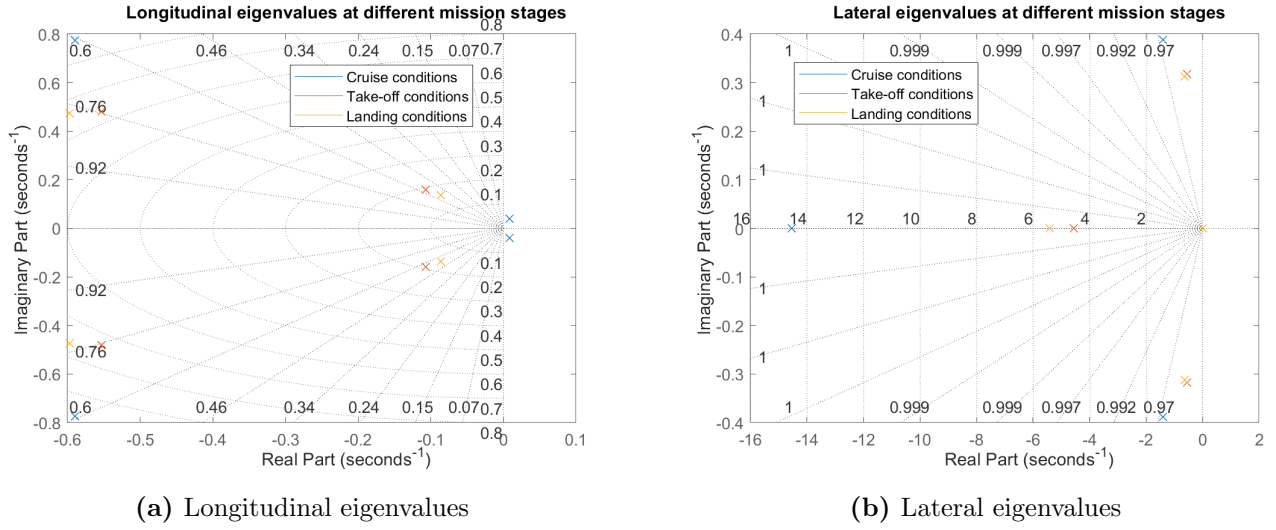


Figure 1: Dynamical modes' eigenvalues

2.4 Lateral motion

Equally, it is possible to introduce the linearised system governing the lateral motion of the aircraft:

$$\begin{bmatrix} m & 0 & 0 & 0 & 0 \\ 0 & I_{xx} & -I_{xz} & 0 & 0 \\ 0 & -I_{xz} & I_{zz} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{V} \\ \dot{p} \\ \dot{r} \\ \dot{\Phi} \\ \dot{\Psi} \end{bmatrix} = \begin{bmatrix} Y_V & Y_p & Y_r - mU_e & mg \cos \Theta_e & 0 \\ L_V & L_p & L_r & 0 & 0 \\ N_V & N_p & N_r & 0 & 0 \\ 0 & 1 & \tan \Theta_e & 0 & 0 \\ 0 & 0 & \sec \Theta_e & 0 & 0 \end{bmatrix} \begin{bmatrix} V \\ p \\ r \\ \Phi \\ \Psi \end{bmatrix} \quad (2)$$

The above ODE contains the lateral stability derivatives as well as the inertial and cruise parameters. After obtaining the numerical values for the lateral derivatives from [9] and using the same inertial properties and cruise values as the ones used for the longitudinal analysis, the system can be solved in a similar fashion and subsequently the eigenvalues characterising the lateral dynamical motion can be found.

	Eigenvalue	ω_n (1/s)	ζ	T_d (s)	Time to double/ half amplitude (s)
Spiral	0.005	-	-	-	159.711
Dutch roll	$-1.400 \pm 0.388i$	1.439	0.963	2.577	0.495
Roll Subsidence	-14.500	-	-	-	0.048

Table 3: Lateral eigenvalues for the cruise segment conditions

Before analysing the aircraft's eigenvalues, it is worth noting that there is one extra eigenvalue that is not displayed in Table 3 or Figure 1b, which is the yaw-independence eigenvalue. This has a value of 0 and it is related to the neutral stability with respect to a change in yaw angle, making the aircraft's dynamic behaviour independent of its heading. Similar to the longitudinal analysis, the following is presented for the cruise condition eigenvalues but the different conditions follow similar trends. The first eigenvalue displayed in Table 3 corresponds to a coupled yawing and rolling motion referred to as the Spiral mode [6]. This eigenvalue is commonly positive (unstable) and has a low magnitude, therefore the amplitude will have a low growth rate (allowing the pilots to correct the motion in time). The Dutch Roll is an oscillatory motion of coupled yawing rolling and sideslipping motion [6]. This mode is typically lightly damped and the period of the mode is typically of the order of 3 - 10 s [5] (in the case of the design presented it is slightly below the range due to the large damping term, which is above the usual trend values for the mode). The final mode is the Roll Subsidence mode, which is dominated by a large rolling moment. This mode is typically highly damped and has a small time to half amplitude. The values obtained for the aircraft generally agree with the predicted trends and therefore validate the latter.

3 Computational modelling of the aircraft

Due to the novel configuration of the aircraft, it was decided to carry out a computational analysis of the aircraft geometry to estimate the stability parameters and aid in the design of the empennage. This chapter will outline the

software selection process, the methodology used to model the aircraft in both software and an outline of the results obtained, evaluating the accuracy of said software and discussing possible sources of disagreement.

3.1 Software selection

A study of the available aircraft design tools was carried out in order to select the optimum software for the objective of this project. The main aim was to find a reliable software (with studies available to validate the accuracy) that could model the correct aircraft geometry (therefore ideally being able to model 3D shapes) that could account for compressibility effects and include a viscous solver (to realistically model the flow characteristics). From the different options considered it was concluded that the combined use of both OpenVSP and flow5 would facilitate iterating over different design choices whilst providing accurate values for the parameters.

OpenVSP is a parametric aircraft design tool that allows for the design of a 3D geometry shape and contains several different levels of analysis, including mass and aerodynamic properties [10]. The aerodynamic solver is able to predict the stability characteristics of the aircraft. The solver, VSPAERO, is a potential flow solver equipped with a VLM (Vortex Lattice Method) and a panel method solver. These two are commonly used aerodynamic solver options. Although it is beyond the scope of the project to explain the theory behind these two methods, it is important to provide a brief discussion of their basis. As explained by Anderson in [11], the VLM is a 2D method consisting of covering the body surface with N panels and placing horseshoe vortices of unknown strength Γ_i and a control point P_i at a certain location of the panel. At any control point P_i the velocity induced by all the vortices can be found by means of the Biot-Savart law. Subsequently, by setting the normal velocity at the point to zero, a system of equations is obtained that can be solved for the unknown value of Γ_i . This method is commonly employed to analyse the flow over finite wings. Panel methods use a similar procedure to that of VLM but do consider thickness and are therefore able to model 3D bodies. Solving the system of equations directly can become computationally expensive. VSPAERO solves the system by means of Preconditioned GMRES, a method published in 1986 by Saad and Schulz, which solves linear systems iteratively and, at every step, minimizes the norm of the residual over a Krylov subspace (further details on the method can be found in [12]). The stability parameters are estimated by finding a set of values describing the model (therefore the model fidelity is very important) and performing six additional tests introducing variations in certain flow parameters to subsequently find the slope [13]. Finally, previous studies have carried out result comparisons with other methods and experimental values and these validate the accuracy of the software and recommend its use for conceptual aircraft design applications [14, 15, 16].

flow5 is a potential flow solver described as the next version of the well-known aircraft design tool xflr5 [17]. The flow solver contains several methods including a non-linear lifting line method, two VLM methods, one volume quad panel method and two triangular panel methods [17]. Additionally, it is possible to model viscosity by means of either interpolation of 2D viscous data or by a viscous loop which ensures that the 2D viscous properties and the 3D inviscid solutions are satisfied [17]. The VLM and panel methods used by the solver are of a similar fashion to those described for the case of OpenVSP and therefore the explanation of the latter will be omitted. As compared to its predecessor, flow5 presents new additions that increase the accuracy and applicability of the software [18] (through the introduction of, for example, the triangle Galerkin methods) and it also provides much lower runtimes for large-sized matrices thanks to its link to Intel's MKL (Math Kernel Library) and multi threading approach [19]. Regarding the stability characteristics, flow5 uses finite differences to compute them as opposed to an explicit method such as the one used in AVL [20]. Finally, studies have been carried out proving the accuracy of the values computed by the software [21]. Of particular interest for the case being studied is the comparison made with a stability analysis carried out in AVL, presented in [20], where the results prove the ability of the software to predict stability characteristics to a certain degree of accuracy. Some large discrepancies were however found in some of the parameters and no explanation was found for the latter, which will be kept in mind when carrying out the analysis of the aircraft. In all, flow5 was deemed as appropriate for the project aims and, in conjunction with OpenVSP, the computational modelling and analysis of the aircraft was carried out.

3.2 Analysis process in OpenVSP

3.2.1 Modelling the aircraft

The first step in the design process was to model the aircraft in OpenVSP. This was carried out whilst being in constant communication with the SET01 (configuration definition) and SET03 (aerodynamic design) sub teams, as they provided the necessary information from the baseline configuration for the modelling of the aircraft. The modelling process was carried out iteratively, introducing the latest changes communicated by the various sub teams and maximising the fidelity of the model before running the analyses. The final model with the final mesh can be seen in Figure 2 and different views of the model can be seen in Figure 5 in Appendix B. The dimensions and parameters are those from the third design iteration of the aircraft and can be found in [22].

The model reproduces the third iteration design and has some geometric points worth outlining. Regarding the wing, it has been designed as an OpenVSP wing member with the specified geometric parameters: including an AR of 20, a wingspan of 45.96 m, a root chord of 2.33 m, a taper ratio of 0.6, a reference area of 108 m², a quarter-chord sweep angle of 15.4 ° and a dihedral angle of -2 °. The aerofoil selected by the SET03 (Aerodynamic design) sub team is the SC(2) - 0714. The latter was not available in the OpenVSP aerofoil database and the coordinates were manually inputted. The fuselage was modelled using the OpenVSP fuselage shape with four distinct sections using

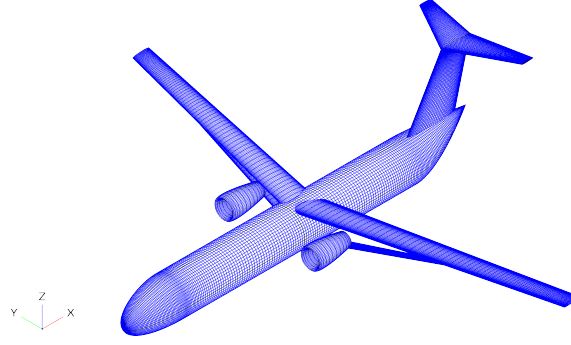


Figure 2: Final model tested in OpenVSP

the dimensions provided by the SET01 (Configuration definition) sub team to obtain the specific fuselage shape. It is a 36.4 m long cylindrical shape with a maximum radius of 4 m and a tailcone length of 7 m. The truss was modelled as a wing with a symmetrical aerofoil (NACA0013) cross section, a span of 27.17 m, an M of 15, a root chord of 1.14 m, a reference area of 24.48 m², a quarter-chord sweep angle of 8° and a dihedral angle of 13.86°. The engines were modelled as a body of revolution, which takes a symmetrical NACA aerofoil cross section and loops it axially to form a tubular shape. Finally, the empennage was modelled separately, as two wings (geometric characteristics can be found in Table 9). The horizontal tailplane was mirrored in the xz plane (default feature for wings) whilst this feature was deactivated for the vertical tailplane. All the elements were positioned based on the information gained from previous iterations and the static stability analysis results (found in [2]).

It is however necessary to note some inaccuracies related to the model. Firstly, for simplicity, the wing-fuselage and truss-fuselage joints were not modelled and were simply attached to each other. These structures will likely have a dampening effect in dynamical motions and an added drag contribution (to be examined in future iterations). Secondly, similar to the previous point, the landing gear housing was not modelled. Thirdly, the wingtip design was refined by the SET03 (Aerodynamic performance) sub team throughout the project and therefore it was decided not to include preliminary designs to avoid inaccuracies stemming from the addition of wingtips. Fourthly, the engine pylons were not modelled because OpenVSP is capable of carrying out the analysis even if two different bodies do not meet and, due to the modelling complexity of the pylon and the probable low effect on the computed values, it was decided to position the engines without them. Finally, the tail of the fuselage was modelled as a point. This is due to errors seen when running the first set of analysis relating to the water-tightness of the mesh. The cross-section of the tail was designed to be circular but it was not possible to implement this. These elements constitute the first set of inaccuracies related to the computational results of the aircraft. It is worth noting that OpenVSP is not a dedicated CAD software, therefore a more detailed model developed in a CAD software should be analysed in future iterations to quantify the errors that result from these inaccuracies in the model.

Despite the mentioned inaccuracies, in order to ensure a high level of fidelity of the model, special care was put when designing the different components. The steps followed to ensure the latter are outlined in [23] and have been developed by the developers of the software. The effects of these improvements were also tested, and a considerable difference in the results was observed. The first element is related to the tessellation of the model, particularly that of lifting surfaces. In order to accurately model the complex flow patterns at the tip and leading and trailing edges, it is recommended to introduce clustering. An example of the implementation of the latter in the horizontal tailplane can be seen in Figure 6 in Appendix B. By introducing clustering, the size of the elements into which the member is divided varies depending on the position on the model. A leading edge clustering and trailing edge clustering of 0.25, a root clustering of 1.5 and a tip clustering of 0.25 were introduced as recommended in [23]. Secondly, when carrying out the mesh convergence study, which will be outlined in the following stage of the report, it was kept in mind that more points do not necessarily produce more accurate results but instead increase the runtime considerably and in occasions worsen the results. Thirdly, it is important to guarantee the mesh is water-tight for the analysis. To this end, the inputted aerofoil properties were slightly modified to satisfy this and the fuselage tail geometry was altered as explained before. Fourthly, in order to maximise the accuracy of the VLM results, straight-sided sections were avoided and, as mentioned, leading edge, trailing edge, tip and root clustering was used. Similarly, to increase the accuracy of the panel method, control surfaces were not introduced on interpolated sections. Having implemented these measures, a high-fidelity model was developed and could start being tested. It is worth noting that, to ensure the quality of the results, as recommended in [23], the results viewer in the GUI was checked after each run to ensure a correct pressure coefficient distribution and the convergence of key aerodynamic parameters such as C_L , C_D and L/D (noting that the best convergence levels after certain wake iterations are of the order of 10^{-5}).

3.2.2 Analysis methodology in VSPAERO

The aerodynamic analysis method employed in the built-in aerodynamic solver in OpenVSP, VSPAERO, to carry out the calculation of the aerodynamic and stability parameters of the aircraft was selected to be the VLM method.

As explained previously in the report, VSPAERO has two solvers, one VLM and one panel method. Both were used for an initial analysis, however the panel method was found to give very large runtimes for simple flow conditions and failed to converge to a reasonable value for different mesh sizes. This is probably due to the more complex geometry and the inability of OpenVSP to carry out a detailed, accurate analysis of the model. This is in agreement to the findings presented in [15], although the results presented in said report are not far from different software. However, due to the time constraints of the project, the need to run a large number of simulations, the accuracy the VLM method presented and the possibility to capture 3D effects with the triangular Galerkin method in flow5, it was decided not to use the panel method in OpenVSP.

The other parameters specified in the VSPAERO GUI include the aircraft reference dimensions (for coefficient calculations, where the wing was selected as the reference lifting surface), the moment reference position (where the center of gravity calculated by OpenVSP was inputted for the reason explained in the section of this report outlining the inertial properties calculations), the flow conditions (where the cruise conditions - $\alpha = -0.7^\circ$, $Ma = 0.75$ & $Re = 1.33e + 07$ - were inputted as obtained from different sub teams), the number of wake iterations - to ensure the convergence of the results - was set to 4 as recommended by the developers in different question forums (the parameters such as C_L , C_D & L/D were checked after each simulation and no issues were found with the convergence of them in any of the runs) and the stability mode was set to steady (to obtain the desired parameters) with the advanced flow conditions setting set, again, to the cruising conditions.

3.2.3 Mesh convergence study

In order to ensure the validity of any computational results it is essential to carry out a mesh convergence study to examine the effect of the mesh refinement on the values computed. There are various approaches that can be taken when carrying out this analysis and there are a large number of factors that can influence the results related to the mesh (although this much depth is out the scope of this analysis). Due to the nature of the values being computed, the approach taken to study the mesh effects was as follows. A base mesh was picked based on initial tests carried out to give a valid result from initial estimates and based on recommended values (seen in Table 10 in Appendix B). Subsequently, a test was ran for different number of elements for each of the components, keeping the others constant with the base mesh size. The quality of the mesh was initially worsened and tests were ran for increasing number of elements. The results obtained for the different components can be seen in Figures 7 - 11 in Appendix B, and the final tessellation values are outlined in Table 4 (where the numbers for the wing, horizontal tailplane and truss are for the whole span). Before outlining the contributions of each component to the accuracy of the results, it is worth mentioning that, generally, an aspect ratio of 1 between tessellation in the x and y directions was selected (noticing that rounded elements such as the fuselage and the engines are taken as flat rectangles with an x side size length of the length of the component and a y side size of the perimeter of the rounded cross section and the vertical tailplane is rotated 90° and projected on the xy plane).

	Fuselage	Wing	Horizontal tailplane	Truss	Vertical tailplane	Each engine
Section tessellation in x direction	57	162	162	53	81	18
Section tessellation in y direction	228	162	162	53	81	29
Total number of elements	12296	26244	26244	11236	26244	522

Table 4: Final tessellation parameters for the OpenVSP model

The mesh convergence figures can be found in Appendix B. Firstly, as it can be seen in Figure 7, the main effects of the different fuselage tessellations can be observed on the y moment coefficient (pitching coefficient). A good convergence is obtained with a total number of 12296 elements and tests with a larger number yielded “nan” or visibly incorrect values and have therefore not been included in the study. The wing effects, as displayed in Figure 8 are visible mainly in the x and z direction forces and in the pitching moment coefficient. Relatively good convergence is obtained with a total number of elements of 26244. A spike in the pitching moment coefficient results is observed for 19044 elements. This is probably due to errors in the software and it is line with the findings presented in [24], therefore this spike was disregarded when evaluating the convergence of the results. The effects of the horizontal tailplane are shown in Figure 9, the main effects are seen in the pitching moment coefficient and a slight spike deviating from the converged result is observed for 19044 elements. However, after carrying out the mesh convergence study in the flow5 software (which will be outlined in the coming sections of the report), where a very large number of elements in the horizontal tailplane were found to be needed to accurately model the flow; and noticing that the spike is not as large as the one for the wing, the largest value of numbers was selected for the tailplane tessellation to avoid inaccuracies. The truss effects are presented in Figure 10. Relatively good convergence is seen for all parameters. However, there is a very large spike in the x force coefficient, that changes the order of magnitude of the value. This was, again, disregarded when selecting the final value. However, it was observed that there was a difference in the value for the pitching moment coefficient and further number of elements were tested. These however yielded unrealistic and nan results, and therefore, to attempt to gain the maximum accuracy possible, the maximum value tested was selected. Finally, the vertical tailplane results are displayed in Figure 11. It was not possible to reach a good convergence of all the values since a larger number of elements yielded nan or inaccurate results. Therefore, due to the slight variation observed in the last parameter in the x force coefficient (suggesting a possible source of

error since the vertical tailplane should not have a large effect on this parameter) and due to the already very large number of elements that the second largest value represents, following the recommendations found in [23], the third largest value was selected, as most other parameters showcased a good convergence (further, the vertical tailplane is not expected to have a large effect on the pitching moment coefficient, suggesting that a further study onto this parameter should be conducted). Finally, no effects were observed when varying the engine tessellation from the default parameters and therefore these values were kept.

3.3 Analysis process in flow5

3.3.1 Modelling the aircraft

The process followed to model the aircraft in flow5 was slightly different to that followed for OpenVSP. In the software it is possible to generate lifting surfaces by imputing the necessary geometrical parameters. The latter are the same ones used to model the wing in OpenVSP. The vertical and horizontal tails can be modelled in a similar fashion. The truss was modelled as a wing, similar to what was done in OpenVSP. Finally, the fuselage was imported as an .stl file from the one generated in OpenVSP. Having modelled all the components, they were positioned following the same dimensions as the ones used in the OpenVSP analysis. A view of the final model with the final tessellation can be seen in Figure 3.

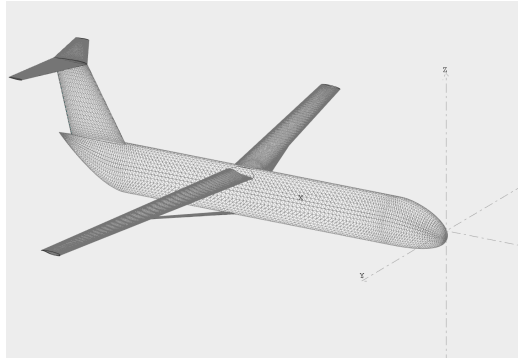


Figure 3: flow5 final model

Similar to what was explained regarding the OpenVSP model the flow5 model contains some inaccuracies. Apart from the ones already mentioned for the OpenVSP model, which are also present in the flow5 model, the engines were not included in flow5. Because of the little effect observed on the VSPAERO simulations and in order to reduce the runtime and complexity of the simulations in flow5, given the time constraints of the project, it was deemed appropriate to carry out this approximation. Further iterations should develop a more realistic model for the simulations.

Finally, the flow5 model was also refined to have the best possible accuracy in the results. A similar philosophy to that followed when finalising the OpenVSP model was followed. The mesh elements were distributed using a *tanh* distribution, as recommended by the developers [25], to accurately model the flow in the different critical sections of the lifting surfaces. Additionally, by modelling the lifting surfaces within the software, a more detailed mesh could be configured, therefore increasing the fidelity of the model. Finally, the aerofoil polars were imported from xflr5 (as flow5, unlike xflr5, does not have access to Xfoil but does have a direct connection with xflr5 to obtain aerofoil data) to be able to carry out viscous analysis. However, the selected aerofoil for the wing was not possible to model through this method due to an error in the software, therefore the polars were requested to the SET03 (Aerodynamic design) sub team.

3.3.2 Analysis methodology in flow5

Unlike VSPAERO, flow5 contains a larger number of solvers and settings that allow for a wide range of simulations to be carried out. Due to the time constraints of the project, it was not possible to carry out an in-depth study of the effects of the different methods and this should be carried out in future iterations. As explained when outlining the software selection, the main aim required from flow5 was to be able to simulate the 3D surfaces and viscous effects. In order to see the importance of the 3D fuselage effects, an analysis was carried out without the fuselage and then compared to the analysis produced by the final mesh. The results are very different, showcasing the importance of a 3D solver, and can be seen in Figure 12 in Appendix C. Therefore, the Galerkin triangular methods were selected as the most appropriate for the analysis. Specifically, the uniform density method was the one chosen as this is the one recommended by the software in the GUI. The lifting surfaces were modelled as thin surfaces due to the complex interactions that could arise from having certain panels overlapping each other. This follows the recommendation given by [23] for better modelling in OpenVSP for panel methods but the concept is applicable in flow5 as well.

A study on the effect of viscosity in the results was carried out but the difference was found to be negligible using both methods available. The xflr5 viscous method was selected and the analyses were carried out with this setting on. The reference dimensions were set to be the main wing ones and the inertial properties calculated by OpenVSP

were used. No control surfaces were added since no control derivatives were computed as part of this analysis (the flow5 predictions for the control derivatives can be found in [8]). The fluid conditions were set to the cruise values ($\rho = 0.365 \text{ kgm}^{-3}$ & $\mu = 3.896\text{e}^{-5} \text{ Nsm}^{-2}$) and the ground effects option was kept off as it was not part of the required analysis. Skin friction drag contributions from the fuselage were accounted for using the fuselage dimensions for the third iteration and solving by the Karman-Schoenherr implicit method (the description of this method is outside the scope of this report but it was deemed as an appropriate approximation for the level of accuracy required at this stage). Finally, no extra drag terms were added apart from the fuselage skin friction (although a model for the engine drag could have been developed and could be accounted for in future iterations) and the wake settings were left to the default ones (5 wake panels per wake column and a total length of 30 times the reference chord of the wing, with an x -progression factor of 1.1).

The above describes the different settings available in the GUI for any kind of analysis. In order to validate the accuracy of the stability results, initial basic aerodynamic analyses were carried out to check the trend of the predictions for different α and ensure the model was properly generated.

3.3.3 Mesh convergence study

A mesh convergence study was also carried out in flow5. The method used is of the same fashion as the one used in OpenVSP. A base mesh size was picked based on default settings and reasonable preliminary results (seen in Table 11 in Appendix C) and the mesh settings of each component were varied studying the effect on the results. The figures outlining the findings are shown in Appendix C. It is important to note that throughout the mesh study, the $\tanh$ distribution of elements was used and a study on the fuselage effects was not carried out since the fuselage mesh was inherent to the .stl file generated by OpenVSP and when this was varied within flow5 errors appeared. Further iterations should carry out studies investigating these two elements.

	Wing	Horizontal tailplane	Truss	Vertical tailplane
Number of elements in x direction	55	75	45	25
Number of elements in y direction	55	75	45	25
Total number of elements	12100	22500	8100	2500

Table 5: Final mesh parameters for the flow5 model

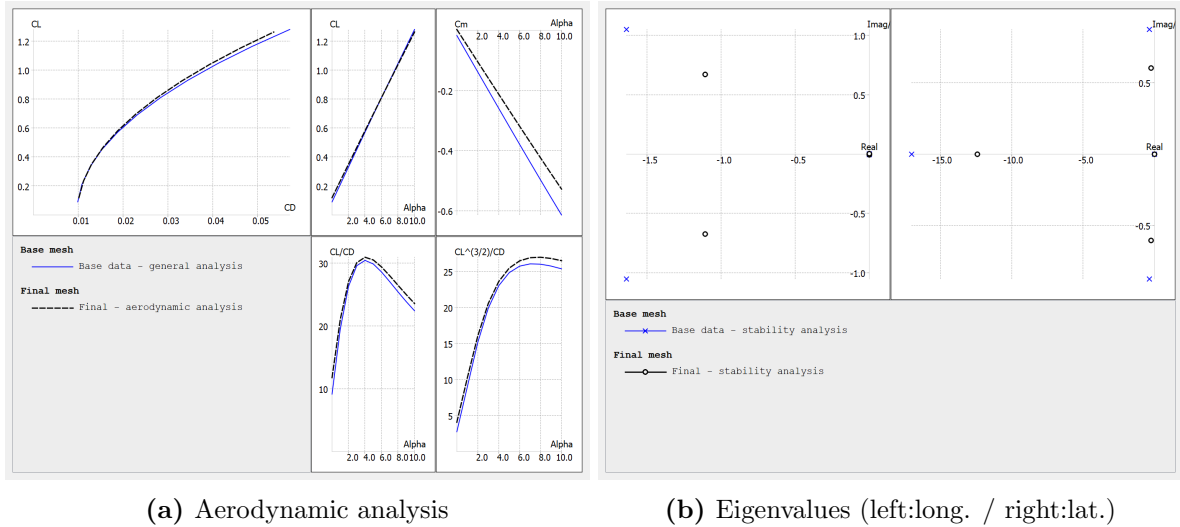


Figure 4: flow5 mesh comparison between base mesh and final mesh

The mesh convergence studies carried out varying the wing mesh can be seen in Figure 13. It is possible to see that a large number of panels is required to accurately capture the flow characteristics and yield converged results. The effects can be clearly seen in all the different aerodynamic analysis curves and in the eigenvalues. The horizontal tailplane analysis was more lengthy (and can be seen in Figure 14), as several different mesh sizes needed to be tested. The effects can mainly be seen in the pitching moment coefficient graph (which is to be expected as it is one of the main effects of the horizontal tailplane), and in the eigenvalues. In order to accurately model the flow a very fine mesh was chosen for the horizontal tailplane (as seen in Figure 3). The effects of the truss can be seen in Figure 15. No major effects can be observed in the aerodynamic curves. However, when zooming into the eigenvalues the convergence required a larger number of points than it required for the aerodynamic curves to converge. Finally, the vertical tailplane analysis is showcased in Figure 16. It was observed that the vertical tailplane mesh had little to no effect in the convergence of results and therefore a low number of elements was selected. The final differences between the base mesh and the fine mesh after the convergence study can be seen in Figure 4, showcasing the importance of the mesh convergence study.

3.4 Evaluation of results

Having modelled the components in both software and carried out the analysis, a comparison of the results produced is now presented. The stability parameters are compared to the ones produced from analytical calculations.

	C_{Y_β}	C_{Y_p}	C_{Y_r}	C_{L_β}	C_{L_p}	C_{L_r}	C_{N_β}	C_{N_p}	C_{N_r}
Analytical	-0.8027	-0.1256	0.4972	-0.0455	-0.8866	0.1583	0.1440	-0.0800	-0.1622
OpenVSP	-0.5810	0.0428	0.4170	-0.0942	-1.0080	0.3144	0.0119	-0.0323	-0.1494
flow5	-0.6947	-0.0597	0.4878	-0.0961	-0.6284	0.0503	0.1533	-0.0245	-0.1095

Table 6: Lateral derivatives comparison

	Phugoid	SPPO	Spiral	Dutch Roll	Roll Subsidence
Analytical	0.0080 ± 0.0402i	-0.5900 ± 0.7920i	0.0048	-1.4000 ± 0.3880i	-14.5000
flow5	-0.0001 ± 0.0300i	-1.1000 ± 4.2300i	-0.0006	-0.2410 ± 3.7900i	-12.4000
X-Plane	-0.0136 ± 0.1640i	-0.6180 ± 1.9700i	0.0050	-0.2220 ± 1.5400i	-6.7900

Table 7: Eigenvalue comparison (with the changes outlined in [2] applied)

The lateral derivatives are compared in Table 6 with those outlined in [9], the eigenvalues are compared with the analytical predictions outlined previously in this report, after some modifications in the derivatives explained in [2] were applied, and with the ones produced by X-plane, outlined in [26]. The comparison between the different computations of the longitudinal stability derivatives is not presented but can be found in [8]. For the latter, generally both software failed to predict the values with accuracy and disagree between them and with the analytical results in the longitudinal cases. Further analyses should be conducted to investigate the reason behind this.

Before evaluating the results presented in the Tables above, it is worth noting that the static margin (characterising longitudinal static stability) was also computed by both software. From the information presented in [2], the expected range of values the static margin may have during the cruise segment is between 4% and 18%, guaranteeing static stability throughout this stage of the mission. The predictions from OpenVSP, when evaluating mid-cruise conditions is 16%, which is slightly far from the middle value of the analytical static margin range but it is within the expected area and at a reasonable distance. This differs from the flow5 prediction, which largely overestimates the static margin in mid-cruise, giving a value of 48%. The discrepancies in these results may arise from similar reasons that explain the large differences observed between the different longitudinal parameters (again, that can be found in [8]).

Regarding the results seen in Table 6, it is possible to see that both software reasonably accurately agree with the analytical values calculated for the lateral stability derivatives. From the OpenVSP perspective, the main differences are observed in C_{Y_p} & C_{N_β} . Both of these derivatives are underpredicted in absolute value and there even is a different sign in the former. The reason for this disagreement is unknown, and further tests should be conducted to identify the most accurate value. Regarding flow5, like OpenVSP, generally agrees quite well with the magnitude and sign with the analytical values as well as the ones predicted by OpenVSP. It is worth noticing how, both software overpredict the magnitude of C_{L_β} , possibly due to the added lift generated by the truss in the crosswind conditions, leading to a larger (negative) rolling motion; and underpredict the magnitude of C_{N_p} , possibly due to the truss impeding the wing from generating a large enough amount of lift to produce the same levels of yawing motion that would be produced without the truss. Finally, it is interesting to note how C_{L_p} & C_{L_r} are predicted quite differently by the different methods in terms of magnitude, which suggests that a further detailed study of these parameters should be conducted.

Table 7 shows the different eigenvalue predictions by the three method used. A detailed analysis of the errors that may arise from X-plane can be found in [26]. In the case of flow5, it generally agrees with the trends but has some noticeable differences. The Phugoid mode's real part is quite different between the three and the damped frequency agrees with the analytical predictions. Both the real and imaginary parts of the SPPO are overpredicted by flow5, which may be related to the finite difference method used by the solver as the other methods seem to agree more closely, therefore making it unlikely that the difference is due to the aircraft configuration. The Spiral is predicted to be stable by flow5 with very low levels of damping, whilst both other methods predict an unstable mode. Again, this may be related to an error in the solver given the good agreement between the other two sources. The two software agree reasonably well with the damping of the Dutch Roll mode, suggesting an added damping effect from the truss that is not predicted by the analytical calculations. However, the frequencies disagree considerably between the three sources, suggesting, again, that a further, more detailed study should be conducted. Finally, flow5 seems to agree reasonably well with the analytical predictions of the Roll Subsidence mode, suggesting possible errors in the X-plane simulations.

The main reason that may have had an impact on the differences outlined above are the difference in mass and inertial properties between the models. As it was explained previously in this report, in order to obtain a clear comparison between the computational predictions and the analytical predictions, computational mass and inertial properties were used in the software and analytical mass and inertial properties were used in the analytical calculations (although the differences between the two were very small). Other reasons that may have had an impact include the aerodynamic values of the lifting surfaces used in the calculations, which the software produces from the simulations and the analytical methods take as input parameters; errors in the model (impeding the software to accurately

compute the parameters); or the inclusion of the truss (which may have an added effect that is not predicted in the analytical formulae used). Finally, the solving methods are not high-fidelity CFD and these calculations should be carried out in future iterations of the design to more accurately predict the stability characteristics of the aircraft.

4 Empennage design

4.1 Empennage configuration decision

In order to reach the emissions reduction goal, novel empennage configurations were considered to substitute the more conventional ones in order to obtain an improved performance. During the design process it was decided to study the possibility of having a Vee-tail configuration as opposed to a T-tail design. Several conceptual studies have been published concluding that this configuration could bring a decrease in fuel consumption [27, 28, 29]. However, due to the overall novel configuration of this aircraft, and the conceptual nature of the studies available, it was decided to carry out a parallel design process of both tail configurations to obtain preliminary performance parameters (mainly weight and drag) and subsequently decide on the best configuration to reach the efficiency goal of the project. The results presented for the T-tail design in the coming subsections correspond to the third (and latest) iteration of the design process and therefore the most updated and optimised design. The results presented for the Vee-tail configuration, that were used to select the best empennage structure for the aircraft, correspond to the second iteration of the aircraft, and therefore the parameters used in the calculation are slightly different to some presented previously in this report and in other reports of the project.

4.2 T-tail design

Due to the high wing configuration of the aircraft, the T-tail shape was selected to be the most appropriate conventional design. This is mainly to avoid the horizontal tailplane being on the wake of the wing, which allows for a larger tail efficiency by reducing the effect of downwash on the latter [30]. Additionally, the configuration allows for a smaller vertical plane due to the end plate effect [30]. There are many different methods to carry out the sizing of the T-tail in the conceptual design stage. The method chosen to follow aims to maximise the efficiency of the tailplane by minimising drag (by reducing the surface area) and weight. The method followed is a variation of the method proposed by Sadraey, that can be found in [30].

4.2.1 Horizontal tailplane sizing

The first step in the process was to select the optimum moment arm as a function of the geometry of the tailcone, aiming to reduce the wetted surface area without leading to a massive increase of the required tailplane size to maintain stability. By analysing the geometric shape of the tailcone of the fuselage, it was possible to derive the following expression for the optimum moment arm, and from it calculate the reference area of the tailplane. It is worth noting that the tailcone shape was varied throughout the design process, and therefore it wasn't possible to derive the most accurate expression for the optimum moment arm. It is suggested that, in future iterations, this is revised to fully optimise the tailplane.

$$l_h = K_c \sqrt{\frac{4\bar{c}S_{ref}\bar{V}_h}{\pi D_{fmax}}} \quad (3) \quad S_h = \frac{\bar{V}_h \bar{c} S_{ref}}{l_h} \quad (4)$$

The non-dimensional parameter $\bar{V}_h$ is a representation of the levels of stability and maneuverability of the aircraft. The higher the parameter, the more longitudinally stable the aircraft will generally be, losing maneuverability (and vice versa) [30]. By analysing similar commercial transport aircraft and of similar tail configurations, the parameter was initially set to 1.1 in the first iterations of the design. However, following the findings from the second iteration, and the fact that due to structural, system locations and stability reasons, the initially obtained moment arm could not be achieved, the parameter was re-examined and set to 1.43 for the third iteration.

The tail geometry was then selected accordingly. The recommended method outlined in [30] was slightly altered in terms of parameter selection due to the novel configuration of this aircraft. The taper ratio is generally set to a similar value to that of the wing, however, following an evaluation of similar aircraft parameters outlined in [31], the taper ratio was reduced slightly from that of the wing (from 0.6 to 0.4). The aspect ratio was not taken to be 2/3 of that of the wing, as that would have yielded an aspect ratio of 13.3, which is unrealistic for a horizontal tailplane. Therefore, following an evaluation of similar aircraft, this parameter was set to 5. The sweep angle was also selected based on similar aircraft. The tail has to be designed so that it does not suffer from compressibility effects, therefore the sweep angle is generally set to that of the wing with a slight increase. However, because the aerofoil selected for the wing is supercritical, the sweep angle selected for the wing was quite low (15.4°). Therefore, the decision of setting the sweep angle 30° was made, and the dihedral angle was set to be the same as the wing's, -2°. The aerofoil design process is outlined in Table 8 (all the aerofoil parameters were obtained from [32]).

The first element to note is that the thickness chosen for the tailplane aerofoil was set to 9%. This is because it is generally recommended for it to be slightly below that of the wing in order to avoid compressibility effects [30]. The different aerofoils considered and their key parameters are shown in Table 8. As suggested by Sadraey in [30], the aerofoil for the tailplane should be able to comfortably produce positive or negative lift, making symmetrical aerofoils preferable. The key parameters to maximise are the $C_{L\alpha}$, $C_{L_{max}}$ and C_L/C_D . All whilst minimising the drag and having a relatively high stall angle of attack. Considering these requirements, it can be seen that, although the

	$C_{L_{max}}$	$C_{L_{\alpha}}$	α_s	$C_{D_{min}}$	C_L/C_D
NACA 0009	1.325	6.170	13	0.0055	115
NACA 63-009	1.150	5.800	11	0.0045	97.2
NACA 64-009	1.175	6.09	11	0.0040	102.9
NACA 65-009	1.100	6	10.5	0.0040	105.5
NACA 66-009	1.075	6.250	10	0.0030	100

Table 8: Tailplane aerofoil selection process

NACA-0009 section has very good $C_{L_{max}}$, C_L/C_D and $C_{L_{\alpha}}$, it has a considerably higher drag than the other sections. Looking at the other options, it was decided that the NACA 66-009 section would provide the combined performance in terms of all the parameters and was therefore selected for both the vertical and the horizontal tailplane.

Future iterations of the design process should look at determining the horizontal tailplane sweep angle in conjunction with the aerofoil design process to ensure the optimum aerodynamic performance is achieved, and a separate aerofoil selection process could be carried out for the vertical tailplane. Finally, the horizontal tailplane setting angle was calculated from the trim analysis outlined in [2]. From those findings, the incidence of the horizontal tailplane was set to be -0.43° .

4.2.2 Vertical tailplane sizing

The vertical tailplane design followed a very similar process to that of the horizontal tailplane. Similarly, for reference, the method suggested by Sadraey [30] was used with some modifications. The vertical tailplane volume coefficient was selected to be 0.08 from evaluating similar aircraft, both in terms of mission and configuration (information found in [30, 31]). As an initial approximation, the vertical tail moment arm was set to that of the horizontal tail. However, following the different design iterations and change of parameters (particularly the sweep angle and the tailplane positioning) this moment arm was reduced to 12.02 m. Subsequently, using a slightly varied form of Eq. 4 (with b instead of $\bar{c}$), the vertical tailplane area was calculated.

The aerofoil chosen was also NACA 66-009 due to the good characteristics for the application. The sweep angle was selected to be the same as the horizontal tailplane, 30° , following an evaluation of similar aircraft parameters found in [30]. The aspect ratio was selected to be 1.55 also from similar aircraft information and the taper ratio was set to that of the wing as it agrees with that of aircraft with similar configurations (0.6).

Two final considerations considered in the T-tail design process were the spin recovery characteristics and the deep stall case of the horizontal tailplane. The former occurs when the rudder is caught in the horizontal tailplane's wake and is blanketed, however this is a problem in a T-tail configuration. Deep stall however is a very dangerous occurrence that consists on the horizontal tailplane being caught in the wake of the wing and therefore impeding the pilot to recover from stall conditions. Due to the high wing positioning this was not considered to be a large issue as it is unlikely the tail will be blanketed by the effects of the wing during stall, however dedicated studies should be conducted to gain better understanding of this factor. In all, the final design parameters of the T-tail can be found in Table 9.

	$\bar{V}$	Moment Arm (m)	Λ ($^\circ$)	Γ ($^\circ$)	S (m ²)	b (m)	$\mathcal{AR}$	$\bar{c}$ (m)	Airfoil
Horizontal tail	1.43	14.90	30	-2	23.80	10.90	5.00	2.18	NACA 66-009
Vertical tail	0.08	12.02	30	-	26.20	6.38	1.55	4.11	NACA 66-009

Table 9: Tailplane geometric parameters

4.3 Vee-tail design

The goal to reduce emissions from the aviation sector has led aircraft designers to consider novel tail configurations. One of the main options proposed is the Vee-tail. Several studies have been conducted studying the efficiency of this tailplane. Most of the studies published, although very preliminary, come to the same conclusion, there is a considerable decrease in wetted surface area but, from weight estimates, there is also a considerable weight increase. Some studies present more optimistic findings [27, 28, 33], with one study presenting a detailed prediction of fuel consumption reductions (up to 1%) [29]. However, the EU-funded NACRE project [34] found that the Vee-tail configuration disadvantages (including stability considerations and overall weight increase) was not a better configuration and was not further investigated. Despite this, it was decided to carry out an initial sizing of the Vee-tail for this aircraft and, from performance estimates, make an educated decision on the optimum empennage structure based on numbers for this particular aircraft.

A handful of methods have been proposed in literature suggesting Vee-tail sizing procedures. Purser and Campbell suggested in their 1945 report - [35] - an analytical method based in the projection of the tailplane into y and z components. However, it was decided to follow the method proposed by Sanchez-Carmona and Cuerno-Rejado in [1], which is a more specific method based on stability requirements for transport aircraft. Some modifications were introduced due to the time constraints of the project and the inherent complexity of the task. The goal was to obtain performance parameters to make an educated decision on whether the Vee-tail would be a better tail for this aircraft.

It is outside of the scope and difficult considering the space constraints of this report to describe the whole method and the assumptions made by the authors. Therefore a brief description will be outlined (the reader should refer to [1]

for a full explanation on the method). The method in [1] takes a reference aircraft (CSR-01, which, in the case of the project would be the aircraft with the T-tail configuration, showcasing the importance of a parallel tailplane design) with all the characteristics known (aerodynamic, stability, mass etc.) and substitute the conventional empennage for a Vee-tail. Subsequently, from the stability and control regulations outlined in CFR-25 [36] four design criteria are extracted and are then implemented to the design by either comparing with the data of the existing aircraft (for the case of stability derivatives), making sure they are satisfied (e.g. the controllability in OEI conditions) or by checking the maximum necessary control surface deflection. Having defined the constraints, all the parameters characterising said constraints are calculated for varying geometries and, from that, a design space is produced, giving the different Vee-tail geometries that would guarantee a stable aircraft.

Several assumptions were made by the authors before carrying out the analysis. Describing all of them is outside of the scope of the analysis presented in the report, therefore only the ones adopted for this analysis will be outlined. Firstly, the control surfaces were modelled as seen in Figure 17 (spanning for 25% of the chord of the Vee-tail). Secondly, the only geometric parameters iterated through were Γ (between 0° and 50°), b (between 13 m and 21 m) and c_r (between 3 m and 4 m). The ranges of values for the parameters are slightly modified from those found in [1] due to the differences in the aircraft configurations. As a slight variation with respect to the method outlined in [1], where the taper ratio was varied too, after analysing the range and the change in the results it produced (minimal) it was decided to set this parameter constant, to 0.3. All the other tailplane parameters were kept constant from the T-tail configuration. The Vee-tail root was positioned in the same place as the vertical tail root. Further studies should be conducting varying the parameters kept constant in this study to analyse possible improvements arising from varying them.

Due to the time constraints of the project, only two of the four analysis presented in [1] were carried out using the second iteration OpenVSP model and VSPAERO. Firstly, the effect of varying b and Γ on C_{M_α} and C_{N_β} (with a fixed 0° control surface deflection) was investigated (presented in Figure 18 in Appendix D), these characterise the stability at cruise. From, Figure 18a, for the aircraft to be at least as stable as the T-tail aircraft the value of the derivative must be situated below the reference value. The opposite is true for Figure 18b, where the value must lie above the reference one. By selecting the smallest possible configuration, a value of $b = 18$ m is chosen. Subsequently, the effects of varying c_r are shown in Figure 19. The same constraints explained above are true for Figures 19a & 19b in Appendix D. From the study, $c_r = 3$ m is selected. It is worth noting that these figures present the trends and values expected from those presented in [1], validating the analysis. The second analysis carried out was the required control surface deflection angle for trim with the preliminary obtained dimensions from the previous analysis. However, this test yielded a required deflection of -41° . This is an unfeasible deflection, which suggests that either there is an error in the analysis method or the Vee-tail needs resizing. This agrees with the preliminary findings presented in [9], where a very large Vee-tail is shown that is needed to guarantee stability for the aircraft based on a preliminary analysis using a different method.

The preliminary Vee-tail design achieved from the first analysis above can be seen in Figure 20 ($\Gamma = 42^\circ$, $b = 18$ m and $c_r = 3$ m). Having obtained the design, an estimation of the drag performance was carried out in OpenVSP. It was found that there was a 35% wetted surface area reduction that accounted for an overall 3% drag reduction. However, due to the geometry of the tail (which would need to be larger to avoid large control surface deflections), having discussed with the team and considering the time constraints, it was decided to optimise the T-tail design for the third iteration and present the findings from the Vee-tail design for a future iteration to consider it. This study was very preliminary and further more detailed studies, particularly on the structure and control coupling effects, should be carried out to quantify a more final performance improvement value.

5 Further work & conclusions

The analysis presented in this report therefore concludes some key elements. Firstly, all the dynamic modes of the aircraft agree with the expected trends for similar conventional aircraft. Secondly, the validity of flow5 and OpenVSP as analysis software has been confirmed with relative satisfaction, noting that the correct use of the software for the correct application is essential, with most of the parameters predict reasonably agreeing with the analytically predicted ones. Finally, the empennage design for the aircraft has been presented alongside the preliminary sizing and performance study of a Vee-tail empennage configuration.

During the report, points to investigate further have been specified and will therefore not be repeated in this section. However, it is important to draw some general future work recommendations. Regarding the dynamical mode predictions, more accurate estimation of inertial properties and stability derivatives should be carried out to gain a deeper insight into the behaviour of the aircraft. Regarding the software predictions, many factors should be improved (as they have already been outlined), but the main recommendation would be to run full CFD simulations on a detailed CAD of the model and perform more in-depth studies of the software used to fully understand the functioning behind the interface. Regarding the empennage design, it is highly recommended to take the Vee-tail method proposed in this report and further investigate the possible benefits in performance this configuration may bring. This should be done in close contact with the structural design team. Regarding the T-tail configuration, more iterations looking to optimise the aerodynamic performance of the tailplane and studies into novel additions to the tail structure, such as double-hinged rudders (as proposed in [34]) should be conducted. In conclusion, apart from correcting the improvements and implementing the suggestions outlined throughout the report, these elements should help develop the aircraft in future iterations.

References

- [1] Sanchez-Carmona A, Cuerno-Rejado C. Vee-tail conceptual design criteria for commercial transport aeroplanes. Chinese Journal of Aeronautics. 2018.
- [2] Kukreti R. Project ZephyrZero:Super-Efficient Transport Group Design Project. Department of Aeronautics, Imperial College London; 2024.
- [3] Etkin B, Reid LD. Dynamics of Flight, Stability and Control. John Wiley and Sons, INC.; 1996.
- [4] Babister AW. Aircraft dynamic stability and response. Pergamon Press; 1980.
- [5] Drela M. Flight Vehicle Aerodynamics. The MIT Press; 2014.
- [6] Levis E. Flight Dynamics and Control Lecture notes. Department of Aeronautics, Imperial College London; 2022.
- [7] Roskam J. Airplane Design, Part V: Component weight estimation. Roskam Aviation and Engineering Corporation; 1989.
- [8] Chen HH. Project ZephyrZero:Super-Efficient Transport Group Design Project. Department of Aeronautics, Imperial College London; 2024.
- [9] Liang P. Project ZephyrZero:Super-Efficient Transport Group Design Project. Department of Aeronautics, Imperial College London; 2024.
- [10] OpenVSP website. Available from: <https://openvsp.org/learn.shtml>.
- [11] Anderson JD. Fundamentals of Aerodynamics. McGraw-Hill education; 2011.
- [12] Youcef S, Schultz MH. GMRES: A Generalized Minimal Residual Algorithm for Solving Nonsymmetric Linear Systems. SIAM. 1986.
- [13] Using VSPAERO. OpenVSP. Available from: <https://openvsp.org/wiki/doku.php?id=vspaerotutorial>.
- [14] Sarode VS. Investigating Aerodynamic Coefficients and Stability Derivatives for Truss-Braced Wing Aircraft Using OpenVSP. Faculty of the Virginia Polytechnic Institute and State University; 2022.
- [15] Cardoso-de Sousa RS, Annes-da Silva RG. NASA-CRM Lateral-directional stability derivatives comparison between semi-empirical, vortex lattice and panel method. Instituto Tecnológico de Aeronáutica; 2021.
- [16] Fanglin Y, Bartasevicius J, Hornung M. Comparing potential flow solvers for Aerodynamic characteristics of the T-Flex UAV. Institute of Aircraft Design, Technical University of Munich; 2021.
- [17] flow5. About flow5. Available from: <https://flow5.tech/index.php/flow5/sample-page/>.
- [18] flow5. 3d Galerkin formulations. Available from: <https://flow5.tech/index.php/flow5/new-features/3d-galerkin-formulations/>.
- [19] flow5. Performance, Intel MKL and multithreading. Available from: https://flow5.tech/docs/flow5_doc/Performance.html.
- [20] flow5. Benchmark against AVL. Available from: https://flow5.tech/docs/flow5_doc/Validation/Stability.html.
- [21] flow5. The Jibe2 experiment - revisited. Available from: https://flow5.tech/docs/flow5_doc/Validation/Jibe2.html.
- [22] Cuntapay C. Project ZephyrZero:Super-Efficient Transport Group Design Project. Department of Aeronautics, Imperial College London; 2024.
- [23] Litherland B. Modelling best-practices for VSPAERO. NASA; 2021.
- [24] Sheridan CND, Pham DDV, Whiteside SKS. Evaluation of VSPAERO Analysis Capabilities for Conceptual Design of Aircraft with Propeller-Blown Wings. NASA; 2021.
- [25] flow5. Modelling - design principles and recommendations. Available from: https://flow5.tech/docs/flow5_doc/Modelling/Modelling.html#wing_strips.
- [26] Yan R. Project ZephyrZero:Super-Efficient Transport Group Design Project. Department of Aeronautics, Imperial College London; 2024.
- [27] Carrier G, Gebhardt L. A Joint DLR-ONERA Contribution to CFD-based Investigations of Unconventional Empennages for Future Civil Transport Aircraft. ONERA.
- [28] Khan MU, et al. Aerodynamic Comparison of Unconventional Aircraft Tail Setup. IEEE. 2019.
- [29] Sanchez-Carmona A, Cuerno-Rejado C. Design Process and Environmental Impact of Unconventional Tail Airliners. Aerospace. 2021.
- [30] Sadraey M. Aircraft design, A Systems Engineering Approach. John Wiley and Sons, INC.; 2013.
- [31] Gudmonsson S. General Aviation Aircraft Design. Elsevier; 2022.
- [32] Abbott IH, Von Doenhoff AE. Theory of wing sections. Dover Publications, INC.; 1959.

- [33] Sanchez-Carmona A, Cuerno-Rejado C, Garcia-Hernandez L. Unconventional tail configurations for transport aircraft. Progress in Flight Physics. 2017.
- [34] Frota J, et al. Final Activity Report, 2005-2010. NACRE; 2010.
- [35] Purser PE, Campbell JP. Experimental verification of a simplified vee-tail theory and analysis of available data on complete models with vee tails, NACA-TR-823. NASA. 1945.
- [36] Code of Federal Regulations. National Archives and Records Administration;. Available from: <https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-25?toc=1>.

Appendix A: Supporting equations

$$I_{xx} = \sum_{i=1}^N m_i ((y_i - y_{cg})^2 + (z_i - z_{cg})^2) \quad (5)$$

$$I_{yy} = \sum_{i=1}^N m_i ((x_i - x_{cg})^2 + (z_i - z_{cg})^2) \quad (6)$$

$$I_{zz} = \sum_{i=1}^N m_i ((y_i - y_{cg})^2 + (x_i - x_{cg})^2) \quad (7)$$

$$I_{xy} = \sum_{i=1}^N m_i (x_i - x_{cg})(z_i - z_{cg}) \quad (8)$$

Appendix B: OpenVSP model supporting data

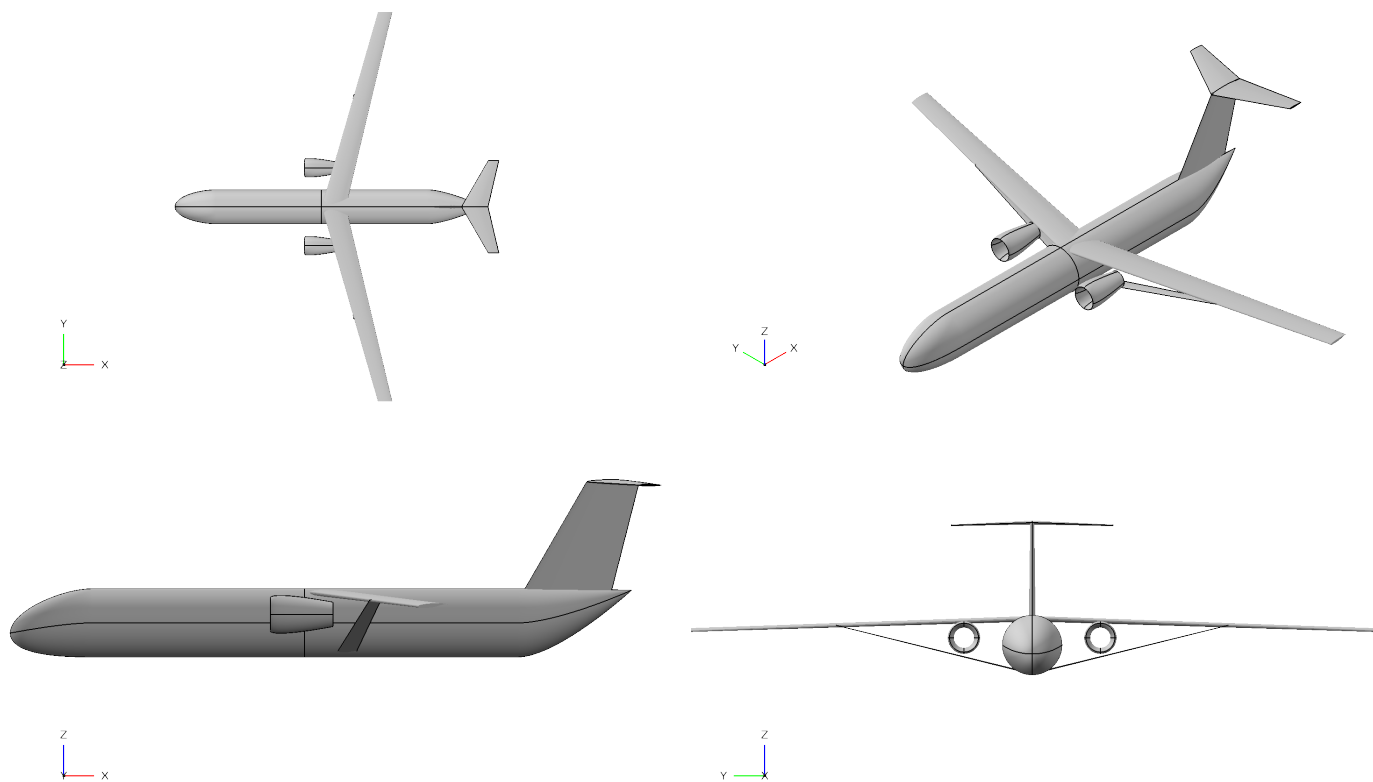


Figure 5: Model tested in OpenVSP

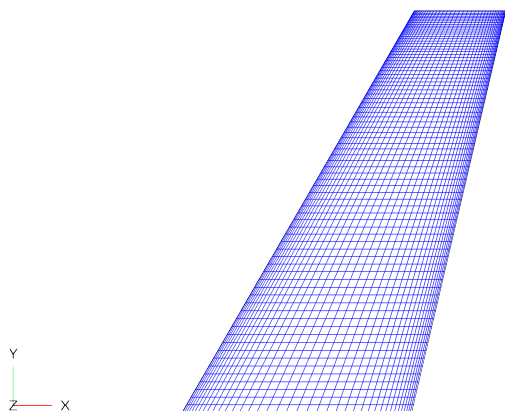


Figure 6: Clustering visualisation

	Fuselage	Wing	Horizontal tailplane	Truss	Vertical tailplane	Each engine
Section tessellation in x direction	49	98	106	34	77	18
Section tessellation in y direction	196	98	106	34	77	29
Total number of elements	9604	9604	11236	1156	5929	522

Table 10: Base tessellation parameters for the OpenVSP model

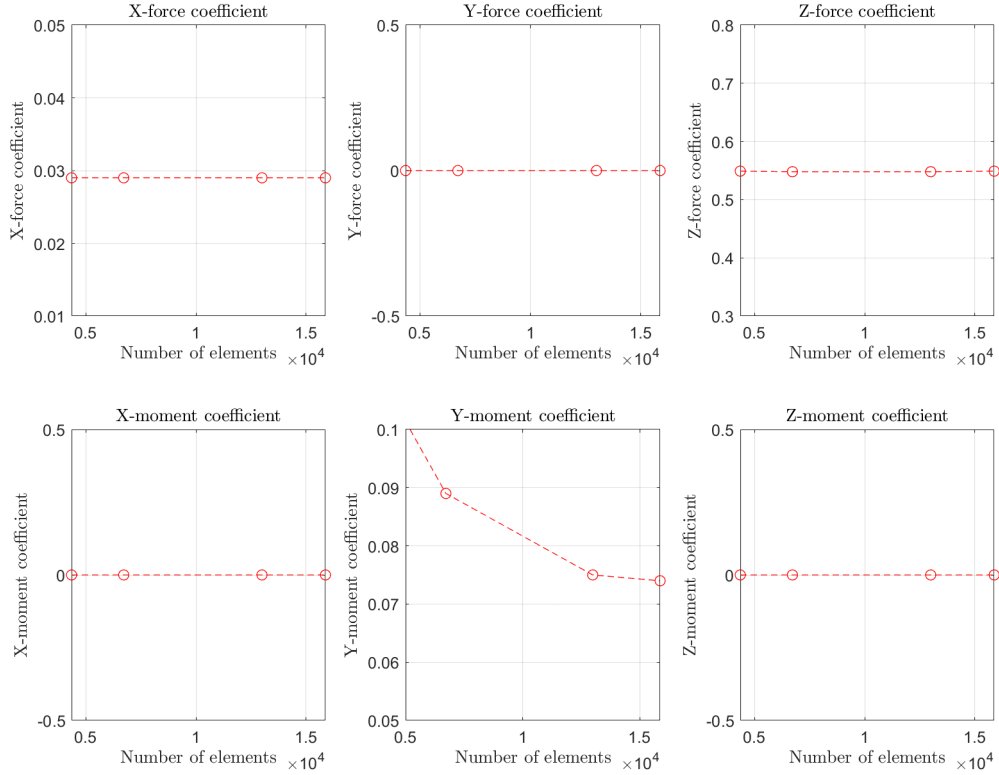


Figure 7: Fuselage number of elements effects on the results convergence in OpenVSP

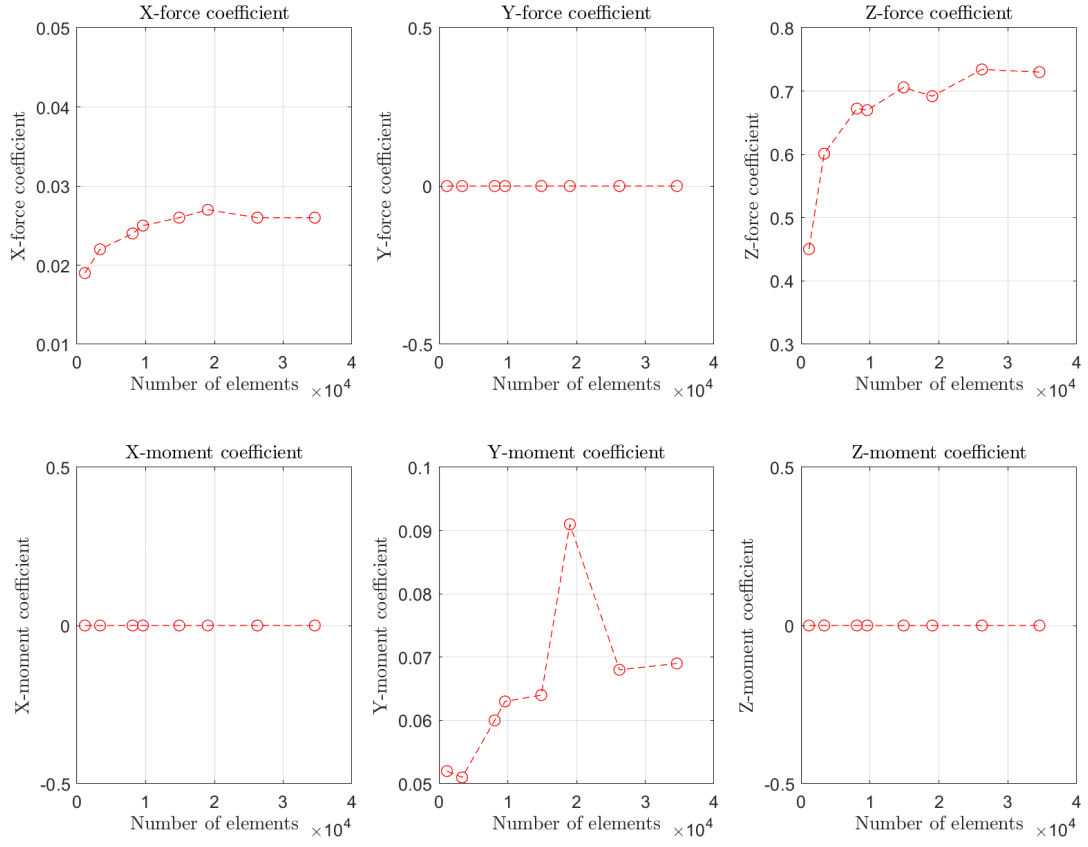


Figure 8: Wing number of elements effects on the results convergence in OpenVSP

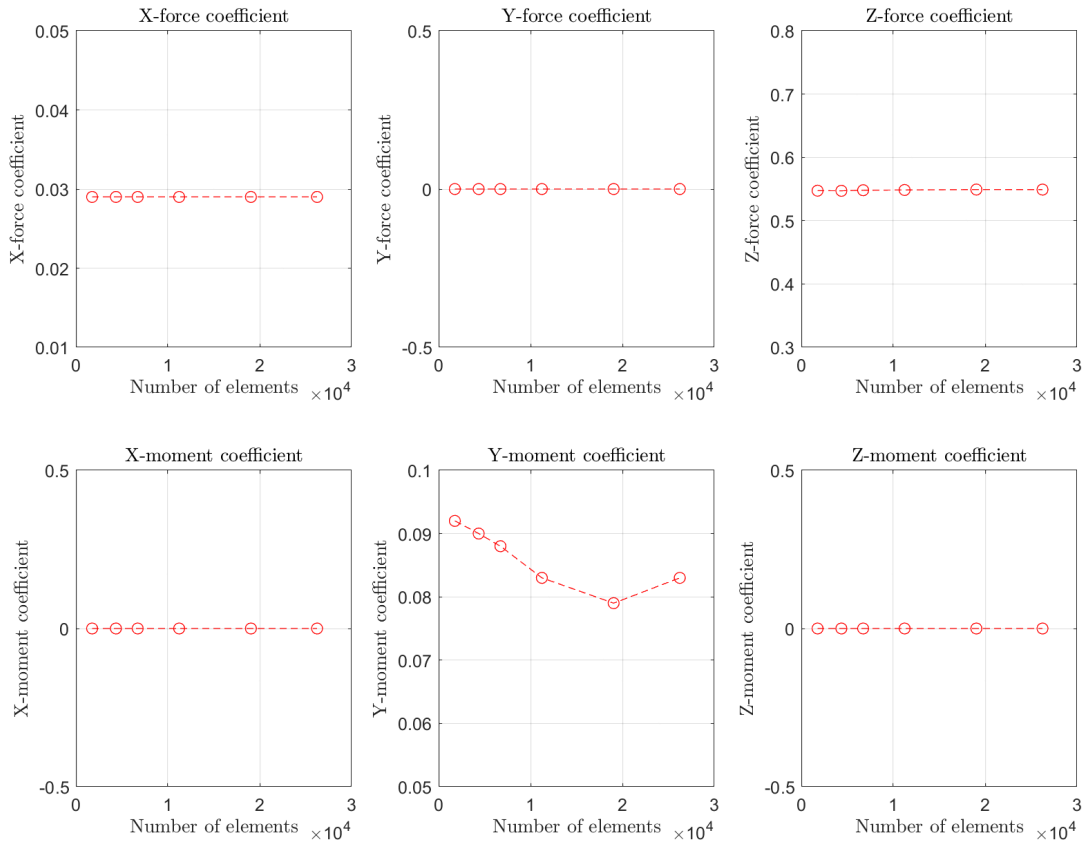


Figure 9: Horizontal tailplane number of elements effects on the results convergence in OpenVSP

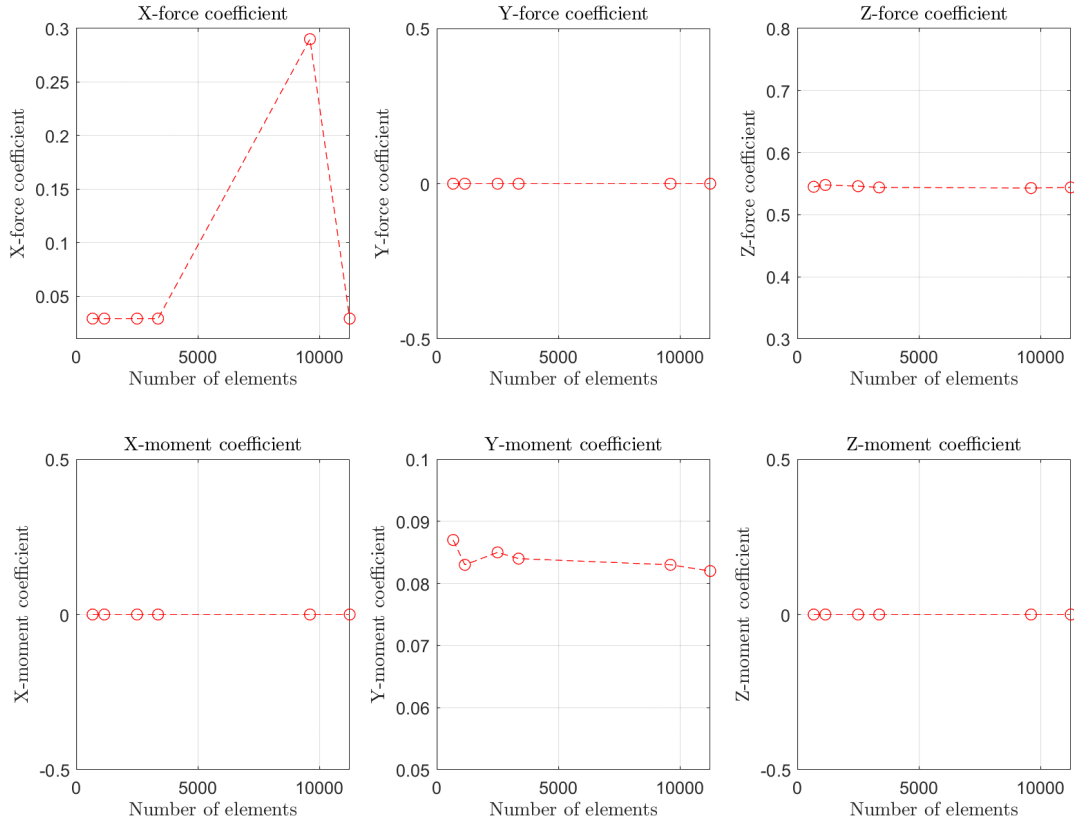


Figure 10: Truss number of elements effects on the results convergence in OpenVSP

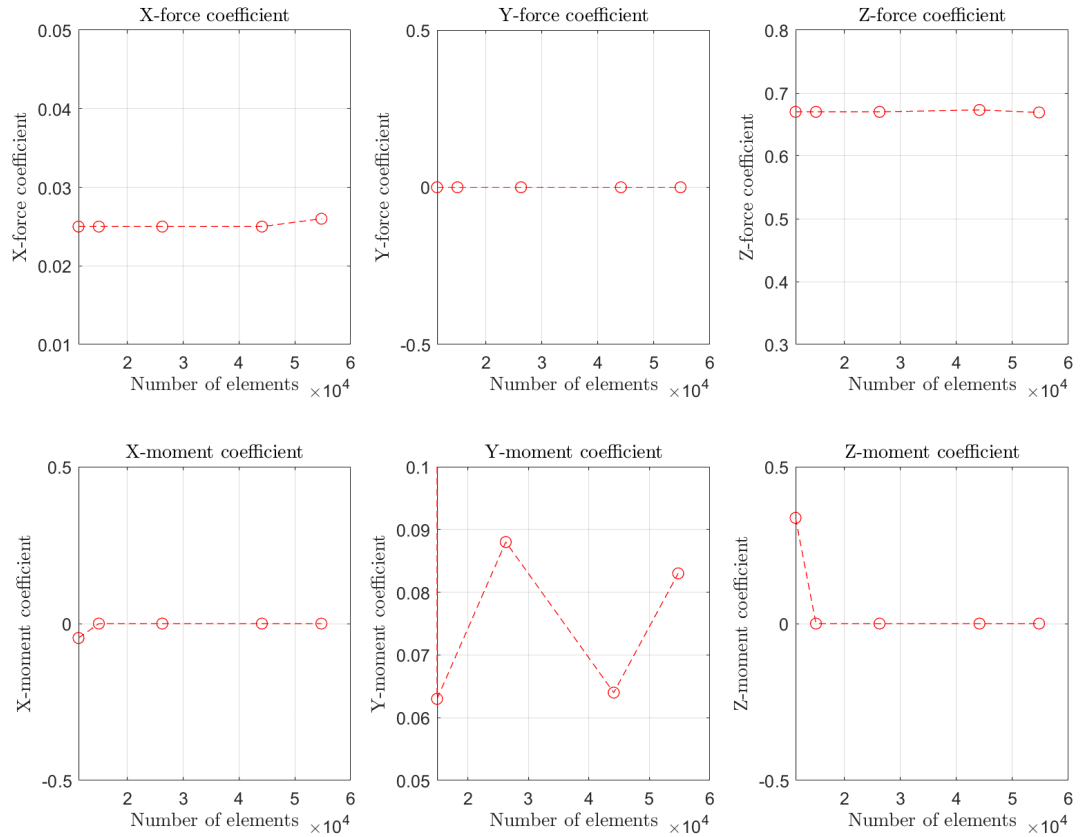


Figure 11: Vertical tailplane number of elements effects on the results convergence in OpenVSP

Appendix C: flow5 model supporting data

	Wing	Horizontal tailplane	Truss	Vertical tailplane
Number of elements in x direction	20	20	20	20
Number of elements in y direction	20	20	20	20
Total number of elements	1600	1600	1600	400

Table 11: Base mesh parameters for the flow5 model

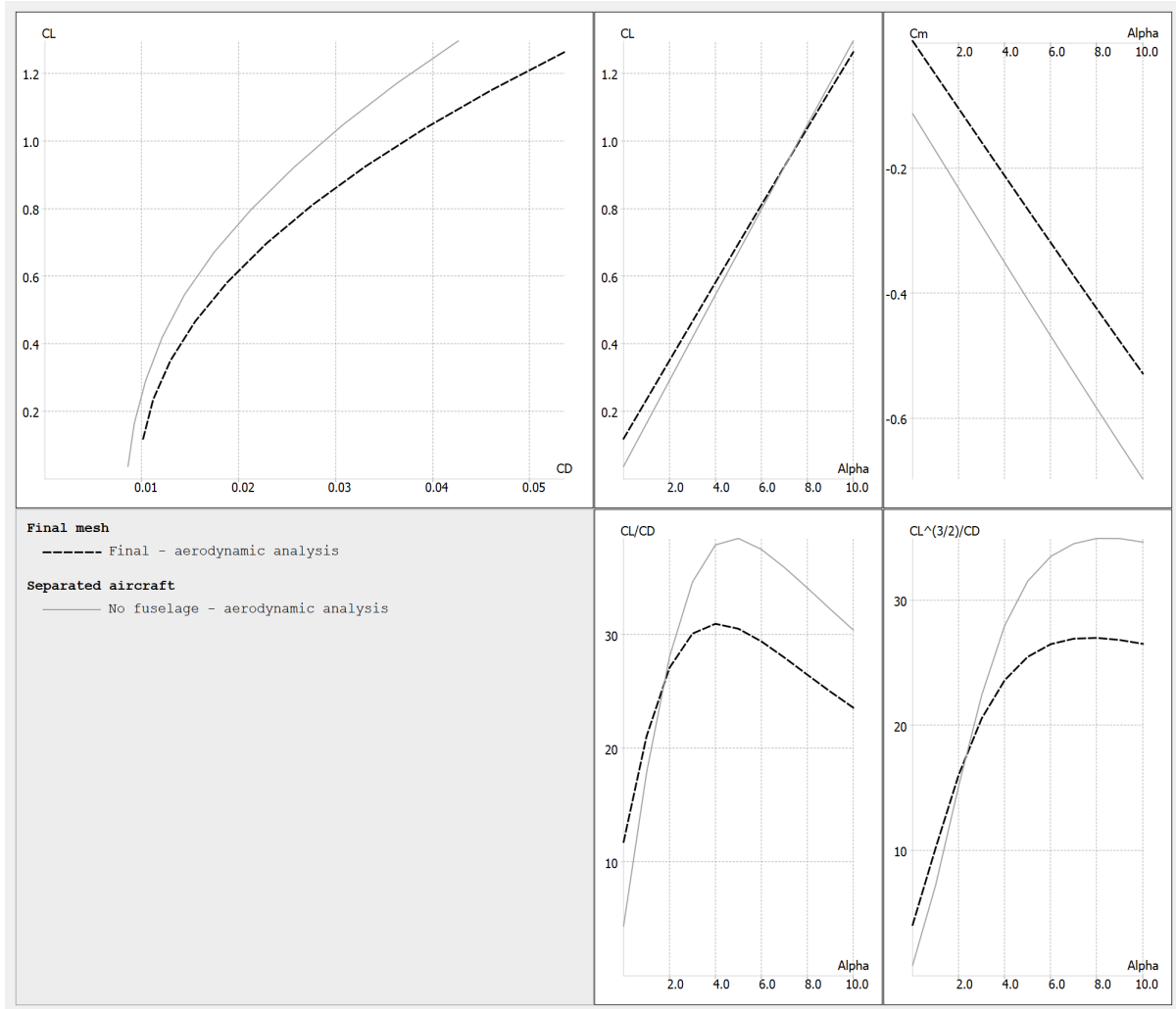
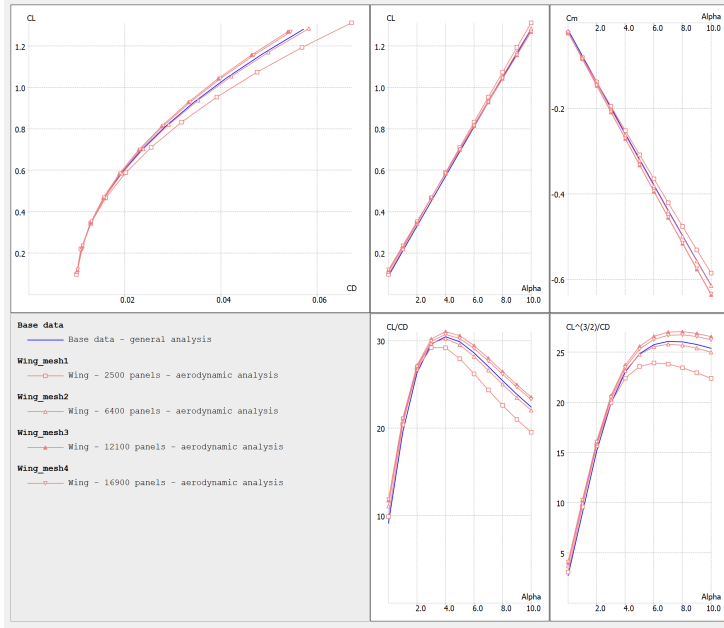
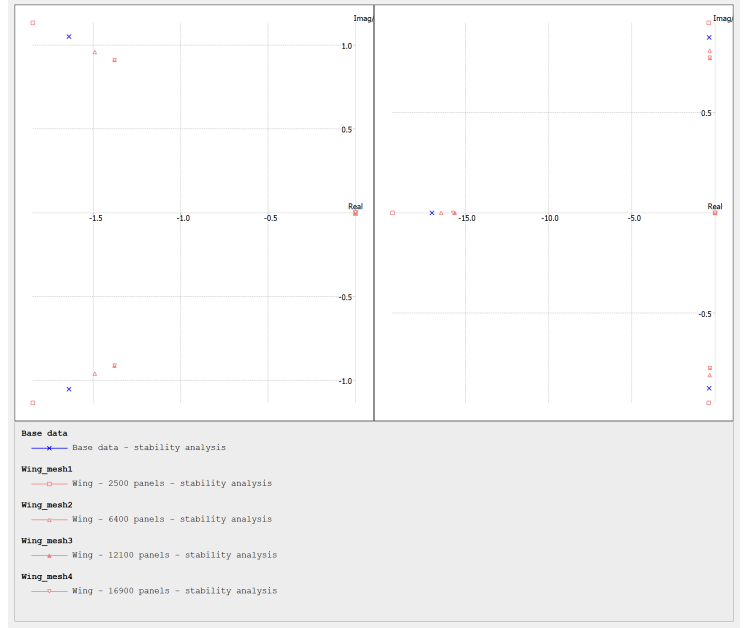


Figure 12: Fuselage effects in aerodynamic calculations in flow5

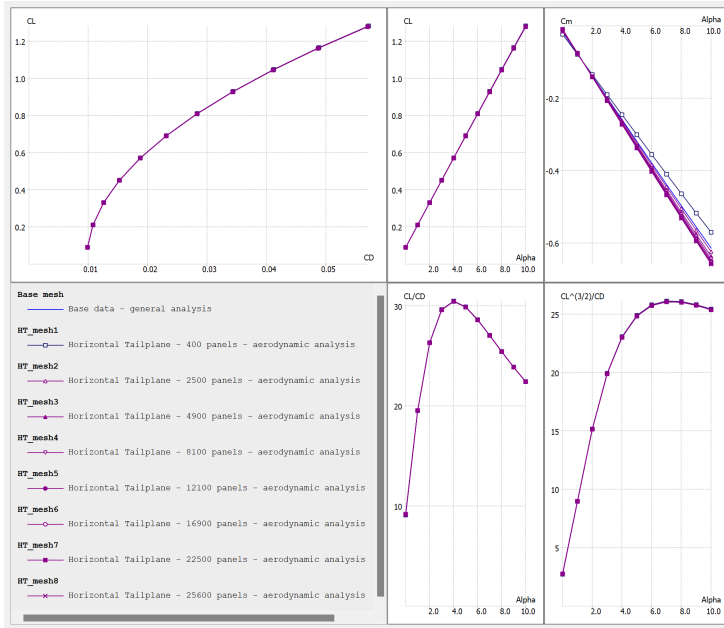


(a) Aerodynamic analysis

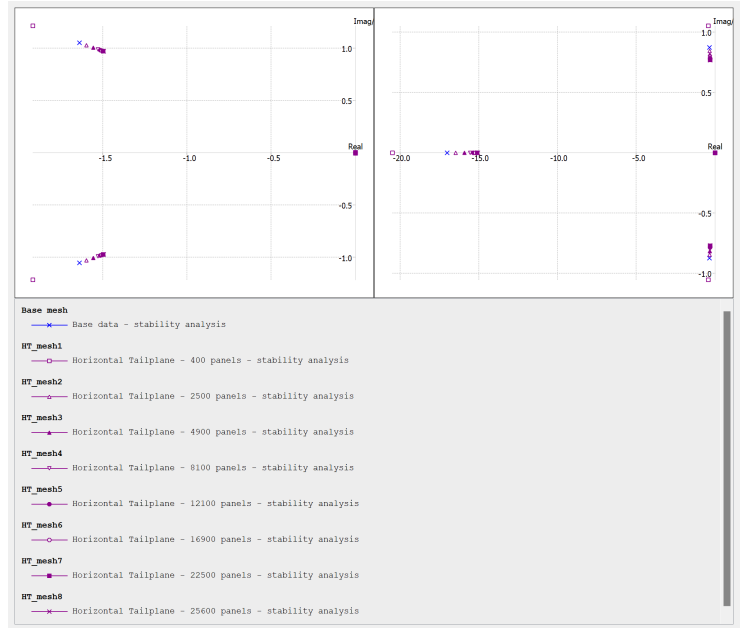


(b) Stability analysis - Left: longitudinal / Right: lateral

Figure 13: flow5 mesh convergence study for different wing tessellation

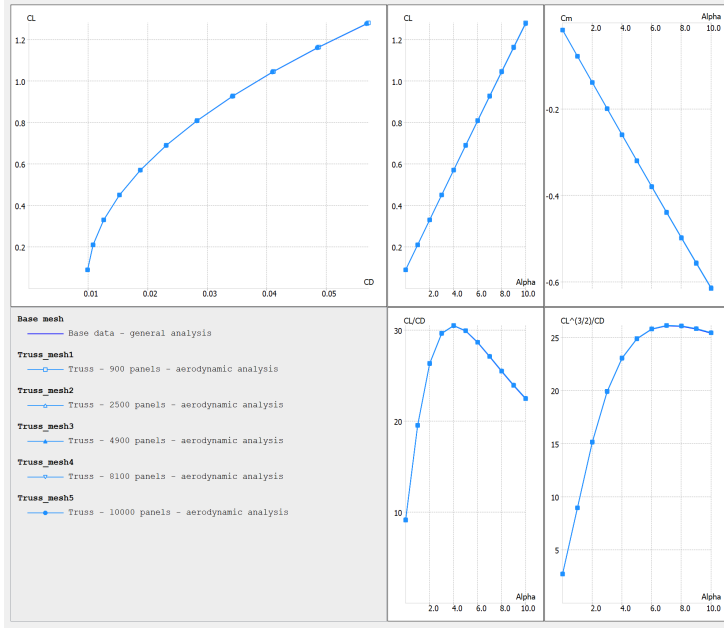


(a) Aerodynamic analysis

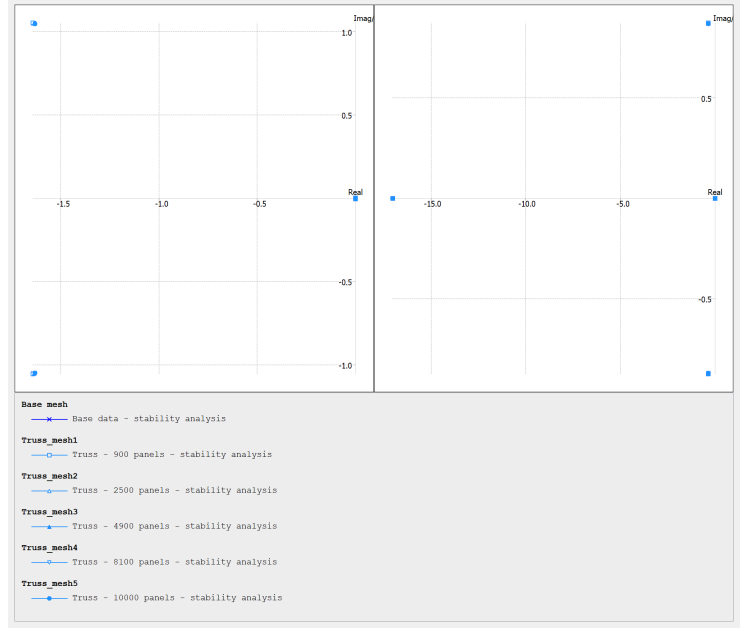


(b) Stability analysis - Left: longitudinal / Right: lateral

Figure 14: flow5 mesh convergence study for different horizontal tailplane tessellation

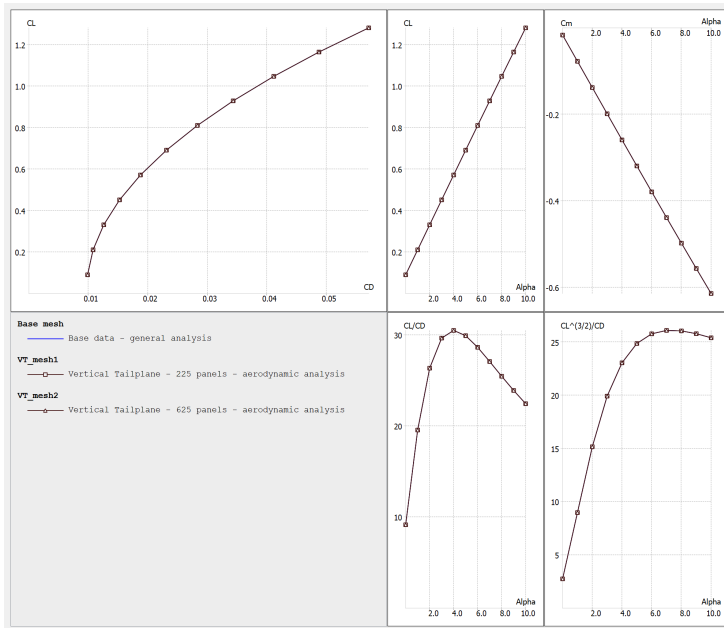


(a) Aerodynamic analysis

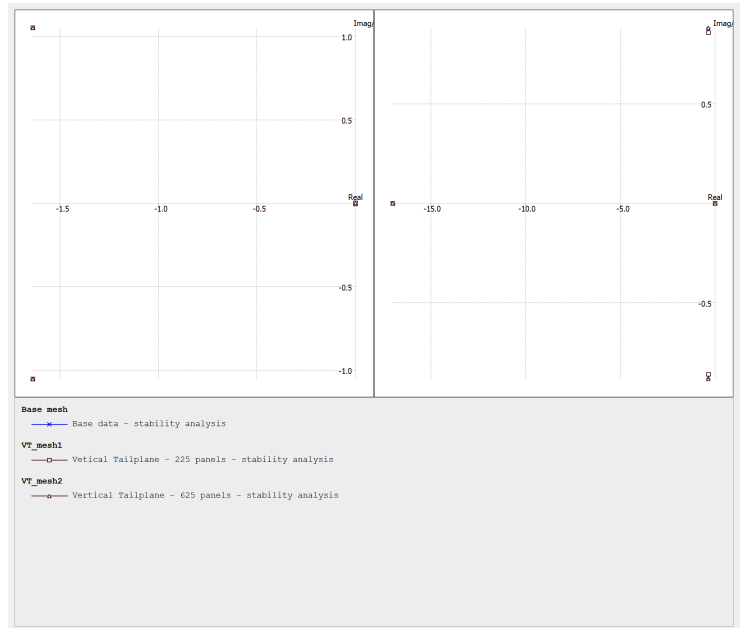


(b) Stability analysis - Left: longitudinal / Right: lateral

Figure 15: flow5 mesh convergence study for different truss tessellation



(a) Aerodynamic analysis



(b) Stability analysis - Left: longitudinal / Right: lateral

Figure 16: flow5 mesh convergence study for different vertical tailplane tessellation

Appendix D: Vee-tail design supporting figures

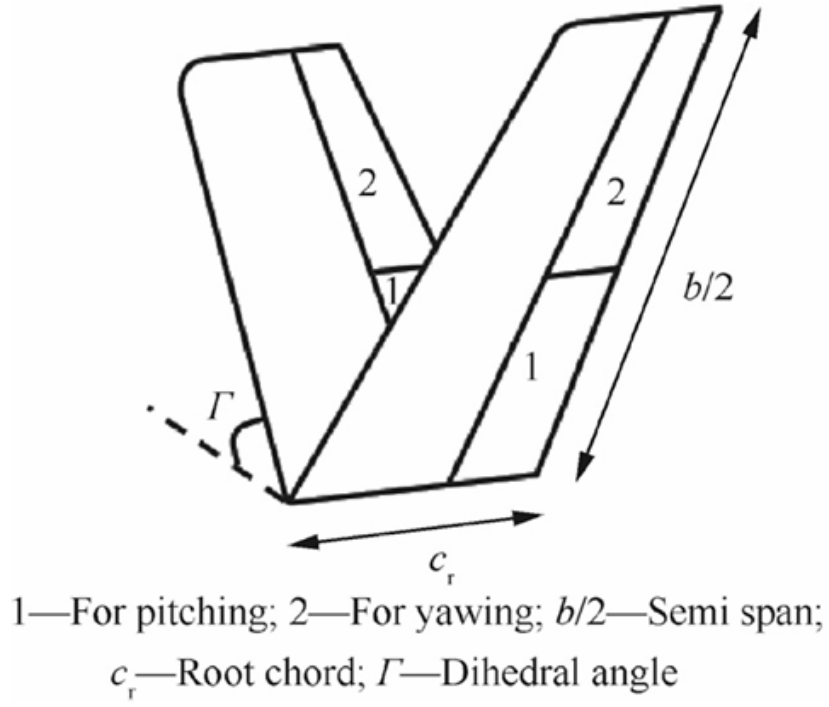
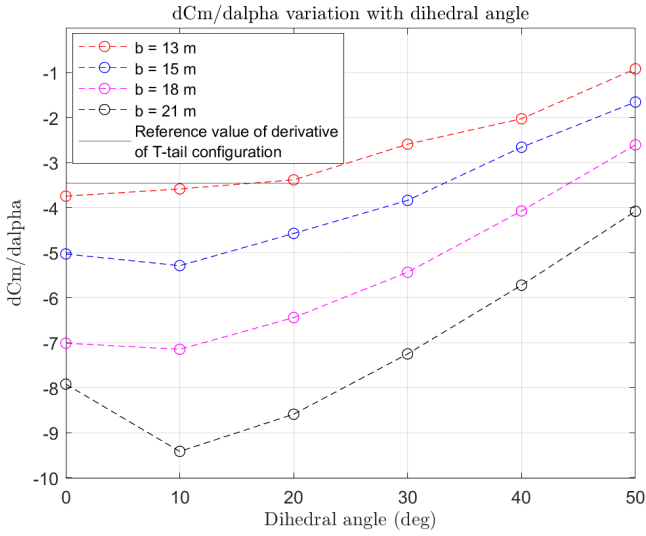
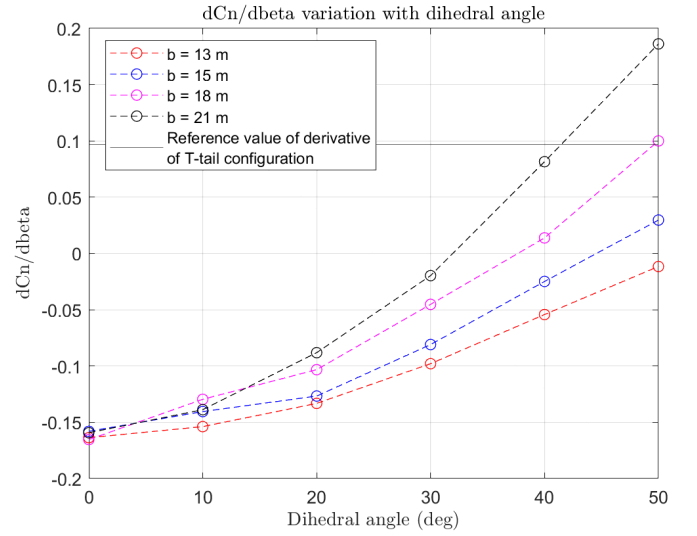


Figure 17: Vee-tail control surface design assumption, reproduced from [1]

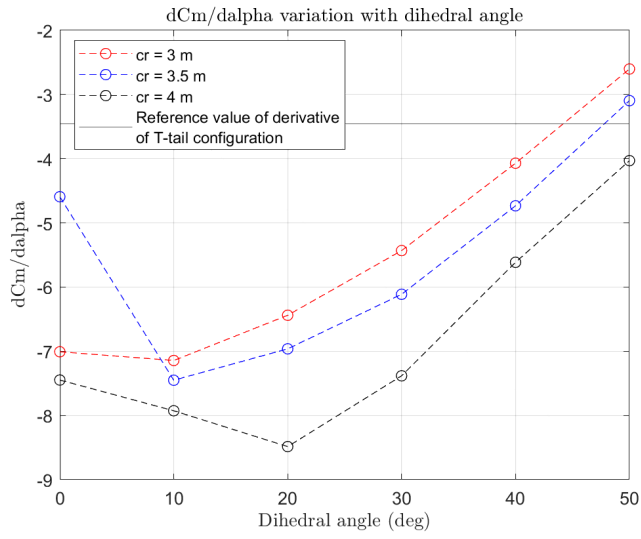


(a) C_{M_α} variation

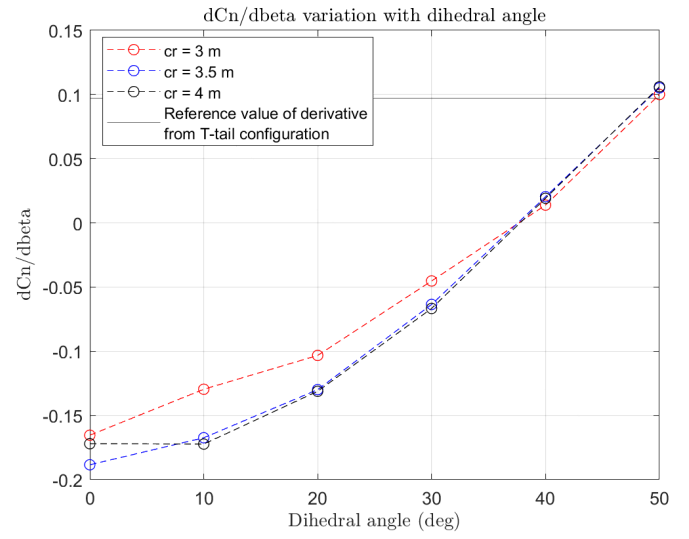


(b) C_{N_β} variation

Figure 18: Variation of stability parameters with Γ & b . Fixed $c_r = 3$ m



(a) C_{M_α} variation



(b) C_{N_β} variation

Figure 19: Variation of stability parameters with Γ & c_r . Fixed $b = 18$ m

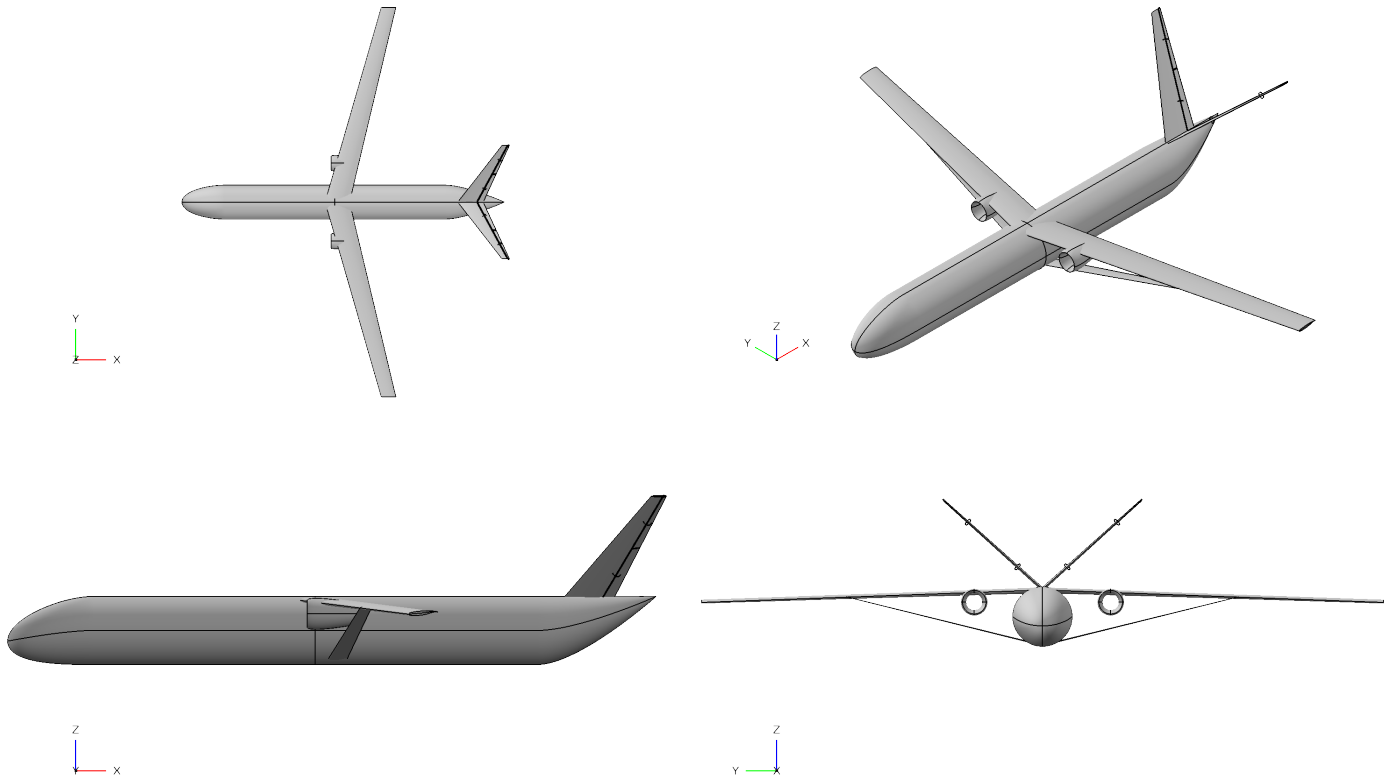


Figure 20: Vee-tail aircraft preliminary design for the second iteration design